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ARTIFICIAL INTELLIGENCE

**Computational Mathematics for Learning &  
Data Analysis (646AA)**

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**Project statement.**

**25. (P)** is so-called *extreme learning*, i.e., a neural network with one hidden layer,  $y = w\sigma(W_1x)$ , where the weight matrix for the hidden layer  $W_1$  is a fixed random matrix,  $\sigma(\cdot)$  is an elementwise activation function of your choice, and the output weight vector  $w$  is chosen by solving a least-squares problem with  $L_1$  regularization  $\arg \min_w f(w)$ , with  $f(w) = \frac{1}{2}\|Xw - y\|_2^2 + \lambda\|w\|_1$ .

**(A1)** is *iteratively reweighted least squares*: i.e., an iteration where you solve at each step the least-squares problem  $w_{k+1} = \arg \min_w f_k(w)$ ,  $f_k(w) = \frac{1}{2}\|Xw - y\|_2^2 + \lambda\|W_k w\|_2^2$  where  $W_k$  is the diagonal matrix with entries  $(W_k)_{ii} = |(w_k)_i|^{-1/2}$ .

**(A2)** is an algorithm of the class of the *deflected subgradient methods*.

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# Chapter 1

## Introduction to the problem

### 1.1 Extreme Learning Machine

Extreme Learning Machine (ELM) [15] is a machine learning model implemented as a neural network with a single hidden layer, in which:

- $\mathbf{W}_1 \in \mathbb{R}^{H \times N}$  is the input-to-hidden weight matrix (where  $H$  is the number of nodes in the hidden layer and  $N$  is the number of input nodes) that is **randomly generated** and **static** (does not change during training) [15].
- $\sigma(\cdot)$  is the element-wise (nonlinear) activation function applied to the  $\mathbf{W}_1 \mathbf{x} \in \mathbb{R}^H$  vector, where  $\mathbf{x} \in \mathbb{R}^N$  is the input vector. The choice of using an activation function for the hidden layer results, under mild assumptions and with high probability, in the hidden layer feature matrix with full rank when  $M \geq H$ , since random projections through a nonlinear function produce linearly independent features.
- $\mathbf{w} \in \mathbb{R}^H$  is the hidden-to-output weight vector.
- $\hat{y} = \mathbf{w}^T \sigma(\mathbf{W}_1 \mathbf{x}) \in \mathbb{R}$  is the prediction of the model.

Once  $\mathbf{W}_1$  is fixed, knowing that  $\mathbf{X}_{\text{origin}} \in \mathbb{R}^{M \times N}$  is the matrix of all the inputs, the hidden-layer activations  $\mathbf{X} = \sigma(\mathbf{X}_{\text{origin}} \mathbf{W}_1^T)$  are constant, and finding  $\mathbf{w}$  reduces to solving the Least-Squares problem described in the following section.

### 1.2 Least-Squares Problem

The definition of the Least Squares Problem [6] in the ELM case is the following:

$$\min_{\mathbf{w}} \|\mathbf{X}\mathbf{w} - \mathbf{y}\|_2^2 \quad (1.1)$$

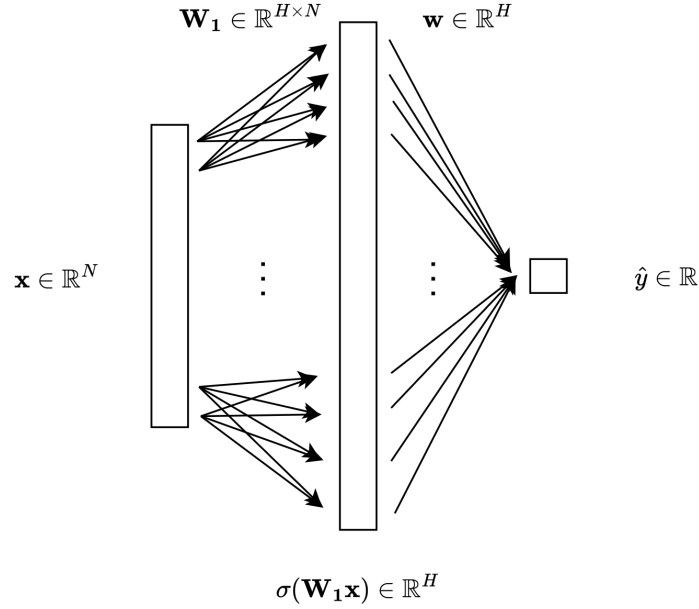


Figure 1.1: Visual representation of an Extreme Learning Machine model.

The Least-Squares problem always admits at least one solution, and the solution is unique if and only if the matrix  $\mathbf{X}$  has full column rank. In that case, we can write the optimization problem as follows:

$$\min_{\mathbf{w}} \frac{1}{2} \mathbf{w}^T \mathbf{X}^T \mathbf{X} \mathbf{w} - \mathbf{y}^T \mathbf{X} \mathbf{w} + \frac{1}{2} \mathbf{y}^T \mathbf{y} \quad (1.2)$$

since the gradient of this function is  $\mathbf{X}^T \mathbf{X} \mathbf{w} - \mathbf{X}^T \mathbf{y}$ , the Hessian matrix is  $\mathbf{X}^T \mathbf{X} \succ 0$ , which means that the function is strictly convex. In this case, the minimum exists and is unique, and can be found solving the equation:

$$\mathbf{X}^T \mathbf{X} \mathbf{w} - \mathbf{X}^T \mathbf{y} = 0$$

where  $\mathbf{X}^T \mathbf{X}$  is square invertible since it's positive definite. The unique solution is  $\mathbf{w}_0 = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$ . However, adding the  $L_1$  regularization term (Section 1.3) destroys the closed-form structure, requiring iterative methods.

## 1.3 Regularization Techniques

### 1.3.1 $L_1$ regularization

When adding the  $L_1$  regularization term to the Least-Squares, the equation is the following:

$$\min_{\mathbf{w}} f(\mathbf{w}) = \frac{1}{2} \|\mathbf{X}\mathbf{w} - \mathbf{y}\|_2^2 + \lambda_{\text{LASSO}} \sum_i^H |w_i| \quad (1.3)$$

This form is particularly nice since it pushes the weights  $w_i \rightarrow 0$ , but the price we have to pay is the non-differentiability of the function where  $w_i = 0$ , which prevents direct use of gradient-based methods.

Because of that, we can use the Deflected Subgradient Methods (2) and the IRLS algorithm (2), which we will see in the following chapters.

## 1.4 Objective Function Analysis

Let's break down the function to minimize in two parts:

$$\min_{\mathbf{w}} f(\mathbf{w}) = \underbrace{\frac{1}{2} \|\mathbf{X}\mathbf{w} - \mathbf{y}\|_2^2}_{g(\mathbf{w})} + \underbrace{\lambda_{\text{LASSO}} \|\mathbf{w}\|_1}_{h(\mathbf{w})}$$

It is useful to analyse the mathematical properties of  $f(\mathbf{w})$ , as they determine which algorithms can be applied.

### 1.4.1 Convexity

The mathematical properties of the objective function  $f(\mathbf{w})$  are heavily dependent on the matrix  $\mathbf{X}$ . As we establish below, a full-rank matrix provides the strictly convex structure necessary to guarantee a unique solution.

**Proposition 1.1.** *If  $\mathbf{X} \in \mathbb{R}^{M \times H}$  has full column rank, then  $f(\mathbf{w}) = \frac{1}{2} \|\mathbf{X}\mathbf{w} - \mathbf{y}\|_2^2 + \lambda_{\text{LASSO}} \|\mathbf{w}\|_1$  is strictly convex and admits a unique global minimizer  $\mathbf{w}^*$  [2, p. 73].*

*Proof. Strict convexity.* Since  $\mathbf{X}$  has full column rank, for any  $\mathbf{v} \neq \mathbf{0}$  we have  $\mathbf{X}\mathbf{v} \neq \mathbf{0}$ , so  $\mathbf{v}^T (\mathbf{X}^T \mathbf{X}) \mathbf{v} = \|\mathbf{X}\mathbf{v}\|_2^2 > 0$ . Hence the Hessian of  $g$  is  $\nabla^2 g = \mathbf{X}^T \mathbf{X} \succ 0$ , making  $g$  strictly convex.  $h$  is convex as a non-negative multiple of a norm. The sum of a strictly convex function and a convex function is strictly convex [12, Slides 21–24], so  $f$  is strictly convex.

**Existence and uniqueness of the minimizer.** Since  $\mathbf{X}^T \mathbf{X} \succ 0$ ,  $g$  is coercive:  $g(\mathbf{w}) \rightarrow +\infty$  as  $\|\mathbf{w}\| \rightarrow \infty$ . Because  $h \geq 0$ ,  $f = g + h$  is also coercive. This means the sub-level set  $\{\mathbf{w} : f(\mathbf{w}) \leq f(\mathbf{0})\}$  is compact, and since  $f$  is continuous, it attains its minimum on this set. Strict convexity then rules out two distinct minimizers: if  $\mathbf{w}_1 \neq \mathbf{w}_2$  both minimized  $f$ , the midpoint  $\frac{\mathbf{w}_1 + \mathbf{w}_2}{2}$  would give a strictly smaller value, a contradiction.  $\square$

### 1.4.2 Non-smoothness

Unlike the quadratic term, the  $L_1$  penalty is **not differentiable** [10, Slide 4] at any point where some component  $w_i = 0$ , so  $f$  is non-smooth. At points where all  $w_i \neq 0$ , the gradient exists and is:

$$\nabla f(\mathbf{w}) = \mathbf{X}^T(\mathbf{X}\mathbf{w} - \mathbf{y}) + \lambda_{\text{LASSO}} \text{sign}(\mathbf{w})$$

where  $\text{sign}(\mathbf{w})$  is applied element-wise. At points where some  $w_i = 0$ , one must reason in terms of *subgradients* (see Chapter 2).

### 1.4.3 Optimality conditions

The optimality condition is expressed via the **subdifferential** [10, Slides 5–6]:  $\mathbf{w}^*$  is a global minimizer for the problem if and only if:

$$0 \in \partial f(\mathbf{w}^*) = \mathbf{X}^T(\mathbf{X}\mathbf{w}^* - \mathbf{y}) + \lambda_{\text{LASSO}} \partial \|\mathbf{w}^*\|_1$$

where the subdifferential of the  $L_1$  norm is given component-wise by:

$$[\partial \|\mathbf{w}\|_1]_i = \begin{cases} \{+1\} & \text{if } w_i > 0, \\ \{-1\} & \text{if } w_i < 0, \\ [-1, +1] & \text{if } w_i = 0. \end{cases}$$

This condition can be written component-wise as: for each  $i = 1, \dots, H$ ,

$$[\mathbf{X}^T(\mathbf{X}\mathbf{w}^* - \mathbf{y})]_i + \lambda_{\text{LASSO}} s_i = 0, \quad s_i \in \partial |w_i^*|. \quad (1.4)$$



## Chapter 2

# Algorithm 1: Iteratively Reweighted Least Squares

Iteratively Reweighted Least Squares (IRLS) [8] is a classical algorithm for solving optimization problems involving non-smooth objective functions of the form of a  $p$ -norm. Its core idea is to handle non-smoothness by solving a sequence of *weighted* least-squares problems, each of which is smooth and admits an efficient closed-form solution. In our setting, IRLS is used to minimize the  $L_1$ -regularized least-squares objective:

$$\min_{\mathbf{w} \in \mathbb{R}^H} f(\mathbf{w}) = \underbrace{\frac{1}{2} \|\mathbf{X}\mathbf{w} - \mathbf{y}\|_2^2}_{g(\mathbf{w})} + \underbrace{\lambda_{\text{LASSO}} \|\mathbf{w}\|_1}_{h(\mathbf{w})} \quad (2.1)$$

where  $\mathbf{X} \in \mathbb{R}^{M \times H}$ ,  $\mathbf{y} \in \mathbb{R}^M$ , and  $\lambda_{\text{LASSO}} > 0$  is the regularization hyperparameter of the original problem.

### 2.1 The core idea: a strict upper bound for the $L_1$ norm

The difficulty in (2.1) is the  $L_1$  term: convex but non-differentiable at every component crossing zero. To replace it with a smooth surrogate we use the elementary inequality  $(|w_i| - |(w_k)_i|)^2 \geq 0$ , which after expansion and division by  $2|(w_k)_i|$  gives the majorization

$$|w_i| \leq \frac{w_i^2}{2|(w_k)_i|} + \frac{|(w_k)_i|}{2}, \quad (w_k)_i \neq 0. \quad (2.2)$$

Summing over  $i = 1, \dots, H$  yields the bound on the  $L_1$  norm

$$\|\mathbf{w}\|_1 \leq \frac{1}{2} \mathbf{w}^T \mathbf{W}_k^T \mathbf{W}_k \mathbf{w} + \frac{1}{2} \|\mathbf{w}_k\|_1 \quad (2.3)$$

where  $\mathbf{W}_k \in \mathbb{R}^{H \times H}$  is a diagonal matrix with entries:

$$(\mathbf{W}_k)_{ii} = |(w_k)_i|^{-1/2} \quad (2.4)$$

This bounds the non-smooth  $L_1$  term using a *smooth, quadratic* expression  $\frac{1}{2} \mathbf{w}^T \mathbf{W}_k^T \mathbf{W}_k \mathbf{w}$ , plus a constant term that depends only on the current iterate  $\mathbf{w}_k$ . Intuitively,  $\mathbf{W}_k^T \mathbf{W}_k$  implements a *personalized penalization* of each component of  $\mathbf{w}$  [5]: components with small magnitude at iteration  $k$  receive a large weight  $|(w_k)_i|^{-1}$  and are strongly pushed toward zero, while components with large magnitude receive a small weight and are left relatively free. This *reweighting* mechanism is what drives IRLS to produce sparse iterates, mimicking the sparsity-inducing effect of the  $L_1$  norm through a sequence of smooth quadratic problems.

**Handling near-zero components.** Equation (2.4) becomes numerically unstable when  $(w_k)_i \approx 0$ , since  $|(w_k)_i|^{-1/2}$  can blow up. To prevent division by (near) zero, a **safety threshold**  $\varepsilon_{\text{thr}} > 0$  is introduced:

$$(\mathbf{W}_k)_{ii} = \left( \max(|(w_k)_i|, \varepsilon_{\text{thr}}) \right)^{-1/2} \quad (2.5)$$

Typical values are  $\varepsilon_{\text{thr}} \in [10^{-6}, 10^{-10}]$ .

## 2.2 The Majorization-Minimization interpretation

IRLS fits naturally inside the Majorization-Minimization (MM) framework [16]. Instead of minimizing  $f$  directly, at each iteration  $k$  we build a *surrogate*  $Q(\mathbf{w}, \mathbf{w}_k)$  that (i) majorizes  $f$ , i.e.  $Q(\mathbf{w}, \mathbf{w}_k) \geq f(\mathbf{w})$  for all  $\mathbf{w}$ ; (ii) touches  $f$  at  $\mathbf{w}_k$ , i.e.  $Q(\mathbf{w}_k, \mathbf{w}_k) = f(\mathbf{w}_k)$ ; and (iii) is smooth, so that it can be minimized in closed form. Plugging the quadratic upper bound (2.3) into (2.1) gives our surrogate,

$$Q(\mathbf{w}, \mathbf{w}_k) = \frac{1}{2} \|\mathbf{X}\mathbf{w} - \mathbf{y}\|_2^2 + \frac{\lambda_{\text{LASSO}}}{2} \|\mathbf{W}_k \mathbf{w}\|_2^2 + \frac{\lambda_{\text{LASSO}}}{2} \|\mathbf{w}_k\|_1, \quad (2.6)$$

which is indeed a quadratic in  $\mathbf{w}$ . The majorization property is built in via (2.2). The touching property follows from a short computation: at  $\mathbf{w} = \mathbf{w}_k$ , each summand of the quadratic penalty becomes  $(\mathbf{W}_k)_{ii}^2 (w_k)_i^2 = (w_k)_i^2 / |(w_k)_i| = |(w_k)_i|$ , so

$$Q(\mathbf{w}_k, \mathbf{w}_k) = \frac{1}{2} \|\mathbf{X}\mathbf{w}_k - \mathbf{y}\|_2^2 + \frac{\lambda_{\text{LASSO}}}{2} \|\mathbf{w}_k\|_1 + \frac{\lambda_{\text{LASSO}}}{2} \|\mathbf{w}_k\|_1 = f(\mathbf{w}_k).$$

Taking  $\mathbf{w}_{k+1} = \arg \min_{\mathbf{w}} Q(\mathbf{w}, \mathbf{w}_k)$ , the two MM properties combined give  $f(\mathbf{w}_{k+1}) \leq Q(\mathbf{w}_{k+1}, \mathbf{w}_k) \leq Q(\mathbf{w}_k, \mathbf{w}_k) = f(\mathbf{w}_k)$ , so the sequence  $\{f(\mathbf{w}_k)\}$  is non-increasing. IRLS thus guarantees a monotone decrease of the objective in exact arithmetic; in floating-point arithmetic the monotonicity is preserved up to the accuracy of the linear solver used for the subproblem.

## 2.3 The weighted least-squares subproblem

To find the next iterate  $\mathbf{w}_{k+1}$ , we must minimize the surrogate function  $Q(\mathbf{w}, \mathbf{w}_k)$  with respect to  $\mathbf{w}$ :

$$\mathbf{w}_{k+1} = \arg \min_{\mathbf{w}} \left( \frac{1}{2} \|\mathbf{X}\mathbf{w} - \mathbf{y}\|_2^2 + \frac{\lambda_{\text{LASSO}}}{2} \|\mathbf{W}_k \mathbf{w}\|_2^2 + \frac{\lambda_{\text{LASSO}}}{2} \|\mathbf{w}_k\|_1 \right)$$

Since the term  $\frac{\lambda_{\text{LASSO}}}{2} \|\mathbf{w}_k\|_1$  is constant with respect to  $\mathbf{w}$ , it does not affect the optimization and can be dropped entirely. The minimization problem simplifies to a standard **weighted least-squares problem with  $L_2$  regularization** (Ridge regression):

$$\mathbf{w}_{k+1} = \arg \min_{\mathbf{w}} \left( \frac{1}{2} \|\mathbf{X}\mathbf{w} - \mathbf{y}\|_2^2 + \lambda_{\text{IRLS}} \|\mathbf{W}_k \mathbf{w}\|_2^2 \right) \quad (2.7)$$

provided that we strictly define the IRLS regularization parameter as:

$$\lambda_{\text{IRLS}} = \frac{1}{2} \lambda_{\text{LASSO}} \quad (2.8)$$

### Relation to the LASSO regularization parameter.

The factor  $1/2$  in (2.8) is forced by the majorization constant in (2.3). The upper bound on  $\|\mathbf{w}\|_1$  carries a coefficient  $\frac{1}{2}$  in front of the quadratic form  $\mathbf{w}^T \mathbf{W}_k^T \mathbf{W}_k \mathbf{w}$ , so the penalty entering the surrogate (2.6) is  $\frac{\lambda_{\text{LASSO}}}{2} \|\mathbf{W}_k \mathbf{w}\|_2^2$ . Rewriting this as  $\lambda_{\text{IRLS}} \|\mathbf{W}_k \mathbf{w}\|_2^2$  in (2.7) therefore requires  $\lambda_{\text{IRLS}} = \frac{1}{2} \lambda_{\text{LASSO}}$ . This is precisely the factor that makes the fixed point of IRLS coincide with the LASSO optimum, as shown in the proof of Theorem 2.2.

**Proposition 2.1** (Weighted normal equations). *The unique solution of (2.7) satisfies:*

$$(\mathbf{X}^T \mathbf{X} + 2\lambda_{\text{IRLS}} \mathbf{W}_k^T \mathbf{W}_k) \mathbf{w}_{k+1} = \mathbf{X}^T \mathbf{y} \quad (2.9)$$

*Proof.* The gradient of the objective in (2.7) is:

$$\nabla f_k(\mathbf{w}) = \mathbf{X}^T (\mathbf{X}\mathbf{w} - \mathbf{y}) + 2\lambda_{\text{IRLS}} \mathbf{W}_k^T \mathbf{W}_k \mathbf{w}$$

Setting this to zero yields (2.9). The coefficient matrix  $\mathbf{X}^T \mathbf{X} + 2\lambda_{\text{IRLS}} \mathbf{W}_k^T \mathbf{W}_k$  is symmetric positive definite (it is the sum of  $\mathbf{X}^T \mathbf{X} \succeq 0$  and  $2\lambda_{\text{IRLS}} \mathbf{W}_k^T \mathbf{W}_k \succ 0$  since  $\varepsilon_{\text{thr}} > 0$ ), so the system has a unique solution.  $\square$

Since  $\mathbf{W}_k$  is diagonal, defining  $\mathbf{Q}_k = \mathbf{X}^T \mathbf{X} + 2\lambda_{\text{IRLS}} \mathbf{W}_k^T \mathbf{W}_k$  and  $\mathbf{b} = \mathbf{X}^T \mathbf{y}$ , equation (2.9) reduces to:

$$\mathbf{Q}_k \mathbf{w}_{k+1} = \mathbf{b} \quad (2.10)$$

This is an  $H \times H$  symmetric positive definite (SPD) linear system, which can be solved efficiently at each iteration. Note that  $\mathbf{b} = \mathbf{X}^T \mathbf{y}$  can be pre-computed once before the iterative loop, since it does not depend on  $k$ . Similarly,  $\mathbf{A} = \mathbf{X}^T \mathbf{X}$  is pre-computed once, and only the diagonal  $2\lambda_{\text{IRLS}} \mathbf{W}_k^T \mathbf{W}_k$  is updated at each iteration.

**Solving the linear system.** Because  $\mathbf{Q}_k$  is SPD, two efficient methods can be used:

- **Cholesky factorization:** [18] computes  $\mathbf{Q}_k = \mathbf{L}\mathbf{L}^T$  in  $O(H^3/3)$  operations, then solves two triangular systems in  $O(H^2)$ . This is the direct method of choice for moderate-sized problems.
- **Conjugate Gradient (CG):** [18] an iterative method that converges in at most  $H$  steps in exact arithmetic, with convergence rate depending on the condition number  $\kappa(\mathbf{Q}_k)$ . Preferable for large-scale problems where  $H$  is too large for Cholesky.

## 2.4 The complete algorithm

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**Algorithm 1** Iteratively Reweighted Least Squares (IRLS)

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**Require:**  $\mathbf{X} \in \mathbb{R}^{M \times H}$ ,  $\mathbf{y} \in \mathbb{R}^M$ ,  $\lambda_{\text{LASSO}} > 0$ ,  $\varepsilon_{\text{thr}} > 0$ ,  $\varepsilon_{\text{stop}} > 0$ ,  $k_{\text{max}}$

```

1: Pre-compute  $\mathbf{A} \leftarrow \mathbf{X}^T \mathbf{X}$ ,  $\mathbf{b} \leftarrow \mathbf{X}^T \mathbf{y}$ 
2: Define  $\lambda_{\text{IRLS}} \leftarrow \frac{1}{2} \lambda_{\text{LASSO}}$ 
3: Initialize  $\mathbf{w}_0 \leftarrow \mathbf{A}^{-1} \mathbf{b}$  (ordinary least-squares solution)
4: for  $k = 0, 1, 2, \dots, k_{\text{max}}$  do
5:   Compute weights:  $(\mathbf{W}_k)_{ii} \leftarrow (\max(|(w_k)_i|, \varepsilon_{\text{thr}}))^{-1/2}$  for all  $i$ 
6:   Assemble system:  $\mathbf{Q}_k \leftarrow \mathbf{A} + 2\lambda_{\text{IRLS}} \mathbf{W}_k^T \mathbf{W}_k$ 
7:   Solve:  $\mathbf{Q}_k \mathbf{w}_{k+1} = \mathbf{b}$  (via Cholesky or CG)
8:   if  $\|\mathbf{w}_{k+1} - \mathbf{w}_k\| / \max(1, \|\mathbf{w}_k\|) < \varepsilon_{\text{stop}}$  then
9:     break (convergence reached)
10:  end if
11: end for
12: return  $\mathbf{w}_{k+1}$ 

```

---

**Initialization.** We initialize  $\mathbf{w}_0$  with the unregularized least-squares solution  $\mathbf{A}^{-1}\mathbf{b}$ : it is already available (we compute  $\mathbf{A}$  and  $\mathbf{b}$  once at the start anyway), the relative magnitudes of its components are a reasonable proxy for which weights the  $L_1$  penalty will shrink, and no component is exactly zero, so the weights  $(\mathbf{W}_0)_{ii} = |(w_0)_i|^{-1/2}$  are well defined without having to lean on  $\varepsilon_{\text{thr}}$  at the first step.

**Stopping criterion.** The algorithm terminates when the relative change between successive iterates falls below  $\varepsilon_{\text{stop}}$ ,

$$\frac{\|\mathbf{w}_{k+1} - \mathbf{w}_k\|}{\max(1, \|\mathbf{w}_k\|)} < \varepsilon_{\text{stop}},$$

where the  $\max(1, \|\mathbf{w}_k\|)$  normalization makes the test scale-invariant and prevents false triggers when  $\|\mathbf{w}_k\|$  is small.

## 2.5 Convergence analysis

**Monotone decrease.** As derived in Section 2.2 from the MM framework, IRLS monotonically decreases the objective:  $f(\mathbf{w}_{k+1}) \leq f(\mathbf{w}_k)$  for all  $k$ . Since  $f$  is bounded below by zero, the sequence  $\{f(\mathbf{w}_k)\}$  converges.

**Theorem 2.2** (Fixed-point consistency). *If the sequence  $\{\mathbf{w}_k\}$  generated by IRLS converges to a point  $\mathbf{w}^*$  such that  $|(w^*)_i| > \varepsilon_{\text{thr}}$  for all  $i$ , then  $\mathbf{w}^*$  satisfies the optimality conditions of the original problem (2.1).*

*Proof.* At a fixed point  $\mathbf{w}_k = \mathbf{w}_{k+1} = \mathbf{w}^*$ , the weighted normal equations (2.9) give:

$$\mathbf{X}^T(\mathbf{X}\mathbf{w}^* - \mathbf{y}) + 2\lambda_{\text{IRLS}} \mathbf{W}_*^T \mathbf{W}_* \mathbf{w}^* = \mathbf{0}$$

Since  $|(w^*)_i| > \varepsilon_{\text{thr}}$  for all  $i$ , the threshold in (2.5) is inactive and  $(\mathbf{W}_*^T \mathbf{W}_*)_ii = |(w^*)_i|^{-1}$  exactly. The  $i$ -th component of the regularizer term is then:

$$(\mathbf{W}_*^T \mathbf{W}_* \mathbf{w}^*)_i = \frac{(w^*)_i}{|(w^*)_i|} = \text{sign}((w^*)_i)$$

Substituting and using  $\lambda_{\text{IRLS}} = \frac{1}{2}\lambda_{\text{LASSO}}$  from (2.8):

$$\mathbf{X}^T(\mathbf{X}\mathbf{w}^* - \mathbf{y}) + \lambda_{\text{LASSO}} \text{sign}(\mathbf{w}^*) = \mathbf{0}$$

Since  $|(w^*)_i| \neq 0$  for all  $i$ , the  $L_1$  norm is differentiable at  $\mathbf{w}^*$ , so  $\partial\|\mathbf{w}^*\|_1 = \{\text{sign}(\mathbf{w}^*)\}$  [10, Slides 5–6]. By the sum rule for subdifferentials [10, Slide 7], the equation above is exactly  $\mathbf{0} \in \partial f(\mathbf{w}^*)$ , the optimality condition of (2.1) [10, Slides 5–6].  $\square$

**Convergence rate.** In practice, the IRLS scheme used here exhibits linear (geometric) convergence of the iterates: this is consistent with the analysis carried out by Daubechies et al. [8] for the closely related basis-pursuit IRLS algorithm, where local exponential (linear-rate) convergence is established under sparsity and null-space assumptions on the design matrix. The error  $\|\mathbf{w}_k - \mathbf{w}^*\|$  thus shrinks by a constant factor at each iteration, set by the condition number of  $\mathbf{Q}_k = \mathbf{X}^T \mathbf{X} + 2\lambda_{\text{IRLS}} \mathbf{W}_k^T \mathbf{W}_k$ . The extra diagonal contribution  $2\lambda_{\text{IRLS}} \mathbf{W}_k^T \mathbf{W}_k \succ 0$  behaves like a Tikhonov-style preconditioner, and typically leaves  $\mathbf{Q}_k$  better-conditioned than  $\mathbf{X}^T \mathbf{X}$  alone, which is what makes the per-iteration linear solve inexpensive and well-conditioned. We do not claim a formal proof of linear convergence for the ELM LASSO setting; the empirical linear rate observed in Section 5.2 is reported as such, with [8] cited only as the closest available analysis under different (basis-pursuit) hypotheses.

**Verification of hypotheses for the ELM problem.** The two assumptions of Theorem 2.2 are satisfied for (2.1) as follows.

1. *Convergence of the sequence.* We work under the assumption that  $\mathbf{X} = \sigma(\mathbf{X}_{\text{origin}} \mathbf{W}_1^T)$  has full column rank, which is standard for the ELM setting when  $M \geq H$  and  $\mathbf{W}_1$  is drawn at random with a nonlinear  $\sigma$  [15].

Under this assumption  $f$  is strictly convex (Proposition 1.1) and admits a unique minimizer  $\mathbf{w}^*$ . The convergence  $\mathbf{w}_k \rightarrow \mathbf{w}^*$  follows in four steps: (i) IRLS monotonically decreases  $f$  (Section 2.2) and  $f \geq 0$ , so  $\{f(\mathbf{w}_k)\}$  is non-increasing and bounded below, hence it converges; (ii)  $f$  is coercive (Proposition 1.1), so every sublevel set is compact and  $\{\mathbf{w}_k\}$  has at least one accumulation point  $\bar{\mathbf{w}}$ ; (iii) continuity of the surrogate  $Q(\cdot, \mathbf{w}_k)$  and the MM descent property  $Q(\mathbf{w}_{k+1}, \mathbf{w}_k) \leq Q(\mathbf{w}_k, \mathbf{w}_k) = f(\mathbf{w}_k)$  force  $\bar{\mathbf{w}}$  to be a fixed point of the IRLS iteration; (iv) by Theorem 2.2, every such fixed point satisfying  $|(\bar{w})_i| > \varepsilon_{\text{thr}}$  is a global minimizer, so  $\bar{\mathbf{w}} = \mathbf{w}^*$  by uniqueness; since every accumulation point coincides with  $\mathbf{w}^*$ , the full sequence converges.

2. *Non-zero components at convergence.* The hypothesis  $|(w^*)_i| > \varepsilon_{\text{thr}}$  holds for the active (non-zero) components of the sparse solution. Components that collapse to zero within  $\varepsilon_{\text{thr}}$  are effectively frozen by the weight clipping (2.5) and drop out of the optimality condition (1.4) — which is exactly the behaviour we want, because these components correspond to features that the  $L_1$  penalty has marked as inactive.

## 2.6 Computational cost

The per-iteration cost of IRLS breaks down as follows:

- **Weight update  $\mathbf{W}_k$ :**  $O(H)$  — simply iterating over the components of  $\mathbf{w}_k$ .
- **Matrix update  $\mathbf{Q}_k = \mathbf{A} + 2\lambda_{\text{IRLS}}\mathbf{W}_k^T\mathbf{W}_k$ :**  $O(H)$  — since  $\mathbf{A} = \mathbf{X}^T\mathbf{X}$  is pre-computed and only the diagonal changes.
- **Linear system solve  $\mathbf{Q}_k\mathbf{w}_{k+1} = \mathbf{b}$ :**  $O(H^3/3)$  with Cholesky, or  $O(pH^2)$  with CG for  $p$  CG steps.

The dominant cost per iteration is the linear system solve:  $O(H^3)$  with Cholesky. However, the total number of IRLS iterations is typically small (10–50), and the pre-computation of  $\mathbf{A} = \mathbf{X}^T\mathbf{X}$  (costing  $O(MH^2)$ ) is amortized over all iterations. The overall cost is therefore:

$$O(MH^2) + k_{\text{IRLS}} \cdot O(H^3)$$

For problems where  $H \ll M$  (many training samples, moderate hidden layer size), the pre-computation dominates and IRLS is very efficient.

## 2.7 Expected behavior on our problem

From the analysis above we expect the following behaviour on the ELM LASSO problem. The objective  $f(\mathbf{w}_k)$  decreases at every iteration, with no oscillations, which makes IRLS easy to monitor and to debug: a single non-decreasing step is

a red flag. On a semilog plot of  $f(\mathbf{w}_k) - f^*$  against  $k$ , we expect an approximately straight line, with a slope set jointly by the conditioning of  $\mathbf{Q}_k$  and by the safety threshold  $\varepsilon_{\text{thr}}$ .

Sparsity is produced dynamically rather than by an external thresholding step: components  $(w_k)_i$  that start out small get assigned increasingly large weights  $|(w_k)_i|^{-1}$ , which pushes them further toward zero at the next iteration; the stable outcome is a weight vector with many components inside an  $\varepsilon_{\text{thr}}$ -ball of zero. The quantitative trade-offs are then driven by two parameters we chose early on: the regularization strength  $\lambda_{\text{LASSO}}$  (the model parameter), and the safety threshold  $\varepsilon_{\text{thr}}$  (a purely numerical one). Since  $\lambda_{\text{IRLS}} = \frac{1}{2}\lambda_{\text{LASSO}}$  is fixed by the majorization, the only free knob in the model is  $\lambda_{\text{LASSO}}$ : larger values give sparser solutions but a higher training residual, smaller values do the opposite; conditioning of  $\mathbf{Q}_k$  also depends on  $\lambda_{\text{LASSO}}$ , so the convergence speed moves with it. For  $\varepsilon_{\text{thr}}$ , smaller values approximate the true  $L_1$  norm more faithfully and yield sharper sparsity, but pay the price in stability whenever a component drifts very close to zero; a value of order  $10^{-8}$  is a reasonable starting point in this trade-off, and we adopt it as our default to be validated experimentally.

## Chapter 3

# Algorithm 2: Deflected Subgradient Method

In this chapter, we apply the deflected subgradient method with Polyak step and with target level (SGPTL) to the ELM LASSO problem.

### 3.1 Motivation: why the plain subgradient method is inadequate for ELM LASSO

The ELM LASSO problem

$$f(\mathbf{w}) = \frac{1}{2} \|\mathbf{X}\mathbf{w} - \mathbf{y}\|_2^2 + \lambda_{\text{LASSO}} \|\mathbf{w}\|_1, \quad (3.1)$$

is nonsmooth due to the  $\ell_1$  penalty. The plain subgradient method converges on convex nonsmooth problems but has two structural limitations on ELM LASSO. First, the update  $\mathbf{w}_{i+1} = \mathbf{w}_i - \alpha_i \mathbf{g}_i$  has no memory [10, Slide 10] and zig-zags on ill-conditioned  $\mathbf{X}^T \mathbf{X}$ ; while it attains the optimal iteration complexity  $\Omega(L^2/\varepsilon^2)$  [4, Thm. 3.2]<sup>1</sup>, the constant  $L$  depends on  $\|\mathbf{X}\|_2$  and the sublevel-set radius. Second, the diminishing-step condition  $\sum_i \alpha_i = \infty$ ,  $\sum_i \alpha_i^2 < \infty$  [10, Slide 10] forces fast step shrinkage and stalls practical progress. Deflection injects memory into the search direction and addresses both issues.

#### 3.1.1 The deflection mechanism

Deflected subgradient methods [10, Slide 15] replace the raw subgradient with a direction that blends the current subgradient with the previous search direction:

$$\mathbf{d}_i = \gamma_i \mathbf{g}_i + (1 - \gamma_i) \mathbf{d}_{i-1} \quad (i \geq 1), \quad \mathbf{d}_0 = \mathbf{g}_0, \quad (3.2)$$

---

<sup>1</sup>From the Polyak bound  $\bar{f}_k - f^* \leq L \|\mathbf{w}_0 - \mathbf{w}^*\|/\sqrt{k}$  in [4, Thm. 3.2] one gets  $k = O(L^2 \|\mathbf{w}_0 - \mathbf{w}^*\|^2/\varepsilon^2)$ , which we write as  $O(L^2/\varepsilon^2)$  absorbing the initial radius. The form  $O(L/\varepsilon^2)$  that appears in some sources is the same order after rescaling  $\varepsilon$  by  $L$ .



where  $\gamma_i \in (0, 1]$  is the deflection parameter. The update is  $\mathbf{w}_{i+1} = \mathbf{w}_i - \alpha_i \mathbf{d}_i$ ;  $\gamma_i = 1$  recovers plain subgradient,  $\gamma_i < 1$  damps the zig-zag. The method specifies  $\gamma_i$  as the minimiser of  $\|\mathbf{d}_i\|^2$  on  $[0, 1]$  [10, Slide 15], which maximises the Polyak step  $\alpha_i$  since  $\|\mathbf{d}_i\|^2$  is its denominator.

## 3.2 Convergence framework: balancing deflection and stepsize

We use the stepsize-restricted Polyak rule [10, Slide 15]:

$$\alpha_i = \frac{\beta_i (f(\mathbf{w}_i) - f^*)}{\|\mathbf{d}_i\|_2^2}, \quad \beta_i \leq \gamma_i, \quad (3.3)$$

with  $\beta_i = \min(\beta, \gamma_i)$  and  $\beta = 1$  [10, Slide 14]. Since  $f^*$  is unknown, we substitute the target level  $f_{\text{ref}} - \delta$  (Section 3.3).

**Deflection floor**  $\gamma_{\min} > 0$ . Condition (3.5) of [7] requires (when  $\lambda_k \geq 0$ ) the uniform bound  $\beta_k \geq \beta^* > 0$  on the stepsize multiplier — in our notation  $\beta_i \geq \beta^*$  for all  $i$ , not just pointwise positivity. Since  $\beta_i = \min(1, \gamma_i)$ , a uniform  $\gamma_i \geq \gamma_{\min} \in (0, 1]$  supplies  $\beta_i \geq \gamma_{\min}$ , i.e.  $\beta^* = \gamma_{\min}$ . The closed-form greedy (3.4) gives  $\gamma_i \in (0, 1]$  at each step but does not bound the sequence away from zero: when consecutive subgradients align near stationarity, the numerator  $\|\mathbf{d}_{i-1}\|_2^2 - \langle \mathbf{g}_i, \mathbf{d}_{i-1} \rangle$  of  $\gamma^*$  in (3.5) tends to  $0^+$  and  $\gamma_i$  decays, dragging  $\beta_i$  and the Polyak step to zero — the iterate freezes. Projecting onto  $[\gamma_{\min}, 1]$  with  $\gamma_{\min} = 0.05$  supplies the missing uniform bound. A sweep  $\gamma_{\min} \in \{0.05, 0.1, 0.2, 0.5\}$  on the convergence instance gives record gaps in  $[10^{-7}, 10^{-5}]$ , with  $\gamma_{\min} = 0.05$  marginally best (Section 5.1).

### 3.2.1 Optimal deflection parameter

The argmin problem

$$\gamma_i \in \arg \min_{\gamma \in [\gamma_{\min}, 1]} \|\gamma \mathbf{g}_i + (1 - \gamma) \mathbf{d}_{i-1}\|_2^2 \quad (3.4)$$

is quadratic in  $\gamma$ ; expanding  $\|\gamma \mathbf{g}_i + (1 - \gamma) \mathbf{d}_{i-1}\|_2^2$  by bilinearity and taking the vertex of the resulting upward parabola gives the closed form (annotated as such in [10, Slide 15]):

$$\gamma^* = \frac{\|\mathbf{d}_{i-1}\|_2^2 - \langle \mathbf{g}_i, \mathbf{d}_{i-1} \rangle}{\|\mathbf{g}_i - \mathbf{d}_{i-1}\|_2^2}, \quad \gamma_i = \min(1, \max(\gamma_{\min}, \gamma^*)). \quad (3.5)$$

The clip onto  $[\gamma_{\min}, 1]$  instead of  $[0, 1]$  is our addition (Section 3.2); the degenerate case  $\mathbf{g}_i = \mathbf{d}_{i-1}$  has vanishing leading coefficient and gives  $\mathbf{d}_i = \mathbf{d}_{i-1}$  for any  $\gamma$ , so we set  $\gamma_i = 1$  by convention. The first iteration is handled analogously:

with  $\mathbf{d}_{-1} = \mathbf{0}$  the formula (3.5) returns  $\gamma^* = 0$ , which would make  $\mathbf{d}_0 = \mathbf{0}$ ; we set  $\gamma_0 = 1$  so that  $\mathbf{d}_0 = \mathbf{g}_0$ , the plain subgradient first step. In the zig-zag regime  $\mathbf{g}_i \approx -\mathbf{d}_{i-1}$  with  $\|\mathbf{g}_i\| \approx \|\mathbf{d}_{i-1}\|$ , the formula gives  $\gamma^* \approx 1/2$  and the oscillatory component of  $\mathbf{d}_i$  cancels.

### 3.3 The target-level mechanism

The Polyak stepsize [10, Slide 12] requires  $f^*$ . The fundamental inequality [10, Slide 10] is

$$\|\mathbf{w}_{i+1} - \mathbf{w}^*\|_2^2 \leq \|\mathbf{w}_i - \mathbf{w}^*\|_2^2 - 2\alpha_i(f(\mathbf{w}_i) - f^*) + (\alpha_i)^2 \|\mathbf{g}_i\|_2^2, \quad (3.6)$$

minimized at  $\alpha_i^* = (f(\mathbf{w}_i) - f^*) / \|\mathbf{g}_i\|_2^2$  [10, Slide 12]. SGPTL replaces  $f^*$  with the target level  $f_{\text{ref}} - \delta$  [10, Slide 14]:  $f_{\text{ref}}$  is the best value found so far,  $\delta > 0$  an estimate of the residual gap.

A failure mode of this substitution is  $f_{\text{ref}} - \delta < f^*$ , in which case  $f(\mathbf{w}_{i+1}) \geq f^* > f_{\text{ref}} - \delta/2$  for every iterate and the sufficient-descent test  $f(\mathbf{w}_{i+1}) \leq f_{\text{ref}} - \delta/2$  is unsatisfiable:  $f_{\text{ref}}$  never updates and the algorithm silently stalls. The mechanism guards against this through an internal patience counter  $r_i = \sum \alpha_j \|\mathbf{d}_j\|$ , the displacement accumulated since the last  $f_{\text{ref}}$  update. When  $r_i$  exceeds a threshold  $R$  without sufficient descent firing,  $\delta$  is contracted to  $\rho\delta$  with  $\rho \in (0, 1)$  and  $r$  is reset. Repeated contractions drive  $\delta \rightarrow 0$ , eventually restoring  $f_{\text{ref}} - \delta \geq f^*$  and unblocking the sufficient-descent test.

### 3.4 The complete algorithm

Algorithm 2 presents the full Deflected Subgradient Method with target level for ELM LASSO.

**Numerical safeguards.** Three pure floating-point guards lie outside the pseudocode. (i) If  $\|\mathbf{d}_i\|^2 < 10^{-30}$  we terminate. (ii) If  $f(\mathbf{w}_i) - (f_{\text{ref}} - \delta) \leq 0$  we set  $\alpha_i = 0$  (condition (3.5) of [7]) and trigger the sufficient-descent update at  $\mathbf{w}_i$ , since  $f(\mathbf{w}_i) \leq f_{\text{ref}} - \delta/2$ . (iii) If  $\mathbf{w}_{i+1}$  has a non-finite component we skip the iteration without touching  $(\delta, f_{\text{ref}}, r)$ .

#### 3.4.1 Parameter calibration for ELM LASSO

SGPTL has five parameters:  $\beta$ ,  $\gamma_{\min}$ ,  $\delta_0$ ,  $\rho$ ,  $R$ . Each is discussed below with the choice we adopt for ELM LASSO and its rationale.

$\beta = 1$ . [7, (3.1)] admits  $\beta_k \in (0, 1]$ . Substituting the Polyak step  $\alpha_i = \beta(f(\mathbf{w}_i) - f^*) / \|\mathbf{g}_i\|_2^2$  into the fundamental inequality (3.6) gives a per-iteration reduction in  $\|\mathbf{w}_i - \mathbf{w}^*\|_2^2$  proportional to  $\beta(2 - \beta)$ , a downward parabola maximised at  $\beta = 1$ ; the deflected variant of [10, Slide 15] inherits the property. We fix  $\beta = 1$  so that  $\beta_i = \min(\beta, \gamma_i)$  is driven by  $\gamma_i$ .

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**Algorithm 2** Deflected Subgradient with Target Level (SGPTL) for ELM LASSO [10, Slide 15]

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**Require:**  $f$  (ELM LASSO objective);  $i_{\max}$ ;  $\beta \in (0, 1]$ ;  $\delta_0 > 0$ ;  $R > 0$ ;  
 $\rho \in (0, 1)$ ;  $\gamma_{\min} \in (0, 1]$   
1:  $\mathbf{w}_0 \leftarrow (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$  *(same OLS warm start as IRLS)*  
2:  $r \leftarrow 0$ ;  $\delta \leftarrow \delta_0$ ;  $f_{\text{ref}} \leftarrow \bar{f} \leftarrow f(\mathbf{w}_0)$   
3: **for**  $i = 0, 1, \dots, i_{\max}$  **do**  
4:   Compute  $\mathbf{g} \in \partial f(\mathbf{w}_i)$  *(subgradient of LASSO)*  
5:   **if**  $i = 0$  **then** *(no  $\mathbf{d}_{-1}$  available)*  
6:      $\gamma_i \leftarrow 1$ ;  $\mathbf{d}_i \leftarrow \mathbf{g}$   
7:   **else**  
8:     Compute  $\gamma_i$  via (3.5) *(optimal deflection, clipped to  $[\gamma_{\min}, 1]$ )*  
9:      $\mathbf{d}_i \leftarrow \gamma_i \mathbf{g} + (1 - \gamma_i) \mathbf{d}_{i-1}$  *(deflected direction)*  
10:   **end if**  
11:    $\beta_i \leftarrow \min(\beta, \gamma_i)$  *(stepsize-restricted,  $\beta = 1$  default; see §3.2)*  
12:    $\alpha_i \leftarrow \beta_i (f(\mathbf{w}_i) - (f_{\text{ref}} - \delta)) / \|\mathbf{d}_i\|_2^2$  *(Polyak step)*  
13:    $\mathbf{w}_{i+1} \leftarrow \mathbf{w}_i - \alpha_i \mathbf{d}_i$  *(update)*  
14:    $\bar{f} \leftarrow \min(\bar{f}, f(\mathbf{w}_{i+1}))$  *(update record)*  
15:   **if**  $f(\mathbf{w}_{i+1}) \leq f_{\text{ref}} - \delta/2$  **then** *(sufficient descent)*  
16:      $f_{\text{ref}} \leftarrow \bar{f}$ ;  $r \leftarrow 0$   
17:   **else if**  $r > R$  **then** *(travel-distance patience exhausted)*  
18:      $\delta \leftarrow \delta \rho$ ;  $r \leftarrow 0$   
19:   **else**  
20:      $r \leftarrow r + \alpha_i \|\mathbf{d}_i\|_2$   
21:   **end if**  
22: **end for**  
23: **return**  $\mathbf{w}_{\text{best}}$  (iterate achieving  $\bar{f}$ )

---

$\gamma_{\min} = 0.05$ . Condition (3.5) of [7] requires  $\beta^* > 0$  strictly, which the floor realises with  $\beta^* = \gamma_{\min}$ . The value inside  $(0, 1]$  is then a tradeoff: a small  $\gamma_{\min}$  keeps the clip from binding on most steps, so the deflected direction stays close to the unconstrained greedy minimiser  $\gamma^*$  of (3.5) and preserves the locally optimal Polyak step, at the cost of a  $1/\gamma_{\min}$  factor in the rate of Theorem 3.1; a large  $\gamma_{\min}$  shrinks the constant but binds the clip frequently, diluting the memory term in  $\mathbf{d}_i$ . We adopt  $\gamma_{\min} = 0.05$ , the smallest value in the sweep  $\{0.05, 0.1, 0.2, 0.5\}$  of Chapter 5 at which the clip activates only on the near-stationarity degenerate steps documented in Section 3.2; the resulting  $20\times$  rate inflation is moderate.

$\delta_0 = 0.1 f(\mathbf{w}_0)$ . [7, Lemma 3.8] requires only  $\delta_0 > 0$ . The scale must be  $f^*$ -free and problem-dependent: we use  $f(\mathbf{w}_0)$  at the OLS warm start, an upper bound on  $f^*$  and the natural a-priori magnitude of the initial gap. The multiplier  $c$  trades off two failure modes. Small  $c$  places the target  $f_{\text{ref}} - \delta$  close to  $f_{\text{ref}}$ , the sufficient-descent test fires almost immediately and  $\delta$  is barely contracted before reaching the true gap; large  $c$  pushes the target below  $f^*$ , the sufficient-descent test is unreachable and only the patience trigger drives progress. The target-level methods of [3, 13, 1] use moderate values, and  $c = 0.1$  sits inside this range. The sweep  $c \in \{0.01, 0.05, 0.1, 0.5, 1\}$  in Section 5.3 keeps the final gap within one order of magnitude. The same section reports a diagnostic comparison against  $\delta_0 = c f^*$  (not admissible as a tuning target but run once as a sanity check, with  $f^*$  cross-validated against CVXPY interior point); in every regime where SGPTL is in fact making progress, the admissible  $\delta_0 = c f(\mathbf{w}_0)$  and the diagnostic  $\delta_0 = c f^*$  agree within a factor of 2 on the synthetic benchmark (5 seeds), within a factor of 2 on *diabetes*  $H = 50$  at small  $c$ , and within 15% on *diabetes*  $H = 200$ . The  $f^*$ -free choice incurs no measurable cost over the inadmissible oracle on instances where the comparison is informative.

$\rho = 0.7$ . [7, Lemma 3.8] admits any  $\rho \in (0, 1)$ . Tradeoff:  $\rho$  close to 1 contracts  $\delta$  slowly, keeping the target too aggressive for a large fraction of the budget;  $\rho$  close to 0 contracts before the algorithm has had time to exploit each target level. The target-level literature [3, 13, 1], [10, Slide 14] recommends the moderate range  $[0.5, 0.7]$ . A smoke-test sweep at  $\rho \in \{0.3, 0.5, 0.7, 0.8, 0.9, 0.95\}$  on the synthetic instance and on the two real-data ELM matrices places different  $\rho$  as the per-instance best, but only  $\rho = 0.7$  is within a factor of two of the best on every tested instance.

$R = 1$ . [7, Lemma 3.8] admits any  $R > 0$  and gives no specific value. Tradeoff:  $R$  too small contracts  $\delta$  on every minor stagnation, wasting potential progress between target updates;  $R$  too large makes the sufficient-descent test the sole driver and stalls if the target is structurally infeasible. The natural reference is the displacement scale of the problem:  $\|\mathbf{w}_0 - \mathbf{w}^*\|$  is of order 1 on ELM LASSO with column-normalised  $\mathbf{X}$ , and the typical per-iteration step  $\alpha_i \|\mathbf{d}_i\|$  under our other defaults is of order  $10^{-3}$ . We adopt  $R = 1$ : comparable to  $\|\mathbf{w}_0 - \mathbf{w}^*\|$ ,

letting the patience window accumulate  $\sim 10^3$  small steps before contracting. The sweep in Section 5.1.2 reports  $\sim 22$  contractions per 8000-iteration run at  $\rho = 0.7$  (against 0 under the first submission's  $R \approx 894$ , see Section 5.1.2); this matches the cadence Lemma 3.8 prescribes for  $\delta_k \rightarrow 0$  together with  $\sum_k s_k = \infty$ .

### 3.4.2 Starting point and patience-threshold calibration

Algorithm 2 uses the same OLS warm start as IRLS,  $\mathbf{w}_0 = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$ , for an apples-to-apples comparison. With  $R = 1$  and  $\gamma_{\min} = 0.05$  the travel-distance branch fires  $\mathcal{O}(10)$ – $\mathcal{O}(100)$  times per 8000-iteration run, contracting  $\delta$  at the pace the theory expects (Section 5.3).

## 3.5 Application to the ELM LASSO problem

### 3.5.1 Subgradient computation for ELM LASSO

For the ELM LASSO objective, decompose  $f = g + h$  where  $g(\mathbf{w}) = \frac{1}{2} \|\mathbf{X}\mathbf{w} - \mathbf{y}\|_2^2$  (smooth) and  $h(\mathbf{w}) = \lambda_{\text{LASSO}} \|\mathbf{w}\|_1$  (nonsmooth).

By the sum rule for subdifferentials [10, Slide 7]:

$$\partial f(\mathbf{w}) = \nabla g(\mathbf{w}) + \lambda_{\text{LASSO}} \partial \|\mathbf{w}\|_1,$$

where  $\nabla g(\mathbf{w}) = \mathbf{X}^T(\mathbf{X}\mathbf{w} - \mathbf{y})$  and the  $\ell_1$  subdifferential is component-wise (Section 1.4.3):

$$\mathbf{g} = \mathbf{X}^T(\mathbf{X}\mathbf{w} - \mathbf{y}) + \lambda_{\text{LASSO}} \mathbf{s}, \quad s_i \in \partial |w_i|. \quad (3.7)$$

For  $w_i = 0$  we take  $s_i = 0$  via `numpy.sign(0)`, which is admissible (it lies in  $[-1, 1]$ ) so Theorem 3.1 applies. It is not the minimum-norm element; the discrepancy in  $|g_i|$  is at most  $\lambda_{\text{LASSO}}$  per component, inflating  $L$  but not the  $\mathcal{O}(L^2/\varepsilon^2)$  order. The issue is moot under Polyak-step updates from the OLS warm start: exact zeros occur on a set of Lebesgue measure zero.

### 3.5.2 Convergence analysis and hypothesis verification for ELM LASSO

**Theorem 3.1** (Convergence of SGPTL, adapted from [10, Slides 14–15], [7]). *Let  $f : \mathbb{R}^H \rightarrow \mathbb{R}$  be convex with  $f^* = \min_{\mathbf{w}} f(\mathbf{w}) > -\infty$  attained at some  $\mathbf{w}^*$ , and let  $\{\mathbf{w}_i\}$  be the sequence generated by Algorithm 2 (the target-level deflected subgradient method we actually run). Assume the subgradients encountered along the run are uniformly bounded,  $\|\mathbf{g}_i\|_2 \leq L$ , and the stepsize-restricted rule  $\beta_i \leq \gamma_i$  holds with  $\gamma_i \geq \gamma_{\min} > 0$ . Then the record values  $\bar{f}_i = \min_{h \leq i} f(\mathbf{w}_h)$  converge to  $f^*$ , and once the target-level estimate has sharpened to  $f^*$  (i.e.  $\delta_k \rightarrow 0$  and  $f_{\text{ref}}^k \rightarrow f^*$ , which Lemma 3.8 of [7] guarantees) the asymptotic rate matches the classical Polyak bound [4, Thm. 3.2], [10, Slide 13] inflated by the deflection floor:*

$$\bar{f}_k - f^* \leq \frac{L \|\mathbf{w}_0 - \mathbf{w}^*\|_2}{\gamma_{\min} \sqrt{k}}, \quad (3.8)$$

so the worst-case complexity to attain  $\bar{f}_k - f^* \leq \varepsilon$  is  $O(L^2/(\gamma_{\min}^2 \varepsilon^2))$ .

*Proof.* We verify the hypotheses of [7, Theorem 3.7] on Algorithm 2 and read off the conclusions. The notation in [7] maps to ours via  $\sigma_k \leftrightarrow 0$  (exact oracle),  $\alpha_k \leftrightarrow \gamma_i$ ,  $\beta_k \leftrightarrow \beta_i$ ,  $\nu_k \leftrightarrow \alpha_i$ ,  $\mu \leftrightarrow \rho$ ,  $X = \mathbb{R}^H$  (no constraint), and  $\lambda_k \leftrightarrow f(\mathbf{w}_i) - (f_{\text{ref}}^k - \delta_k)$ .

(a) *Deflection-consistency condition (2.13).* In [7] condition (2.13) of Lemma 2.4 reads  $d_k = \tilde{d}_k \Rightarrow v_{k+1} = \tilde{d}_k$ , where  $\tilde{d}_k$  is the unprojected deflected direction and  $\hat{d}_k = -P_{T_k}(-\tilde{d}_k)$  the projected one. This is the hypothesis under which the “crucial” inequality (2.12) holds, which the stepsize-restricted analysis relies on. With  $X = \mathbb{R}^H$  the tangent cone is  $T_k = \mathbb{R}^H$  everywhere, so  $\hat{d}_k = \tilde{d}_k$  and necessarily  $d_k = \tilde{d}_k$  at every iteration; the algorithm sets  $v_{k+1}$  to  $d_k = \tilde{d}_k$  (Algorithm 2 carries no projection step), so (2.13) is satisfied trivially.

(b) *Stepsize condition (3.5):*  $\lambda_k \geq 0 \Rightarrow \alpha_k \geq \beta_k \geq \beta^* > 0$ ,  $\lambda_k < 0 \Rightarrow \alpha_k = 0$ . When  $\lambda_k > 0$ , line 12 of Algorithm 2 sets  $\beta_i = \min(1, \gamma_i) \geq \gamma_{\min}$  via the floor on  $\gamma_i$ , so we may take  $\beta^* = \gamma_{\min}$ ; the inequality  $\alpha_k \geq \beta_k$  in (3.5) is  $\gamma_i \geq \beta_i$ , which is exactly the stepsize-restricted rule. When  $\lambda_k \leq 0$  the numerical safeguard of Section 3.4 skips the step — equivalent to  $\alpha_i = 0$  — and triggers the sufficient-descent update  $f_{\text{ref}}^k \leftarrow \bar{f}$ ,  $r \leftarrow 0$  on  $\mathbf{w}_i$  itself. The skip realises (3.5) verbatim; the additional  $f_{\text{ref}}^k$  update is theory-aligned because  $\lambda_k \leq 0$  means  $f(\mathbf{w}_i) \leq f_{\text{ref}}^k - \delta_k \leq f_{\text{ref}}^k - \delta_k/2$ , which is the sufficient-descent condition of [7, Lemma 3.8] evaluated at  $\mathbf{w}_i$  rather than at  $\mathbf{w}_{i+1}$ .

(c) *Target-level threshold condition (3.26):* requires either  $f_{\text{ref}}^\infty = f^* = -\infty$  or  $\liminf_k \delta_k = 0$  together with the series condition (3.25),  $\sum_k \lambda_k / \|\mathbf{d}_k\|^2 = \infty$  (equivalent to  $\sum_k s_k = \infty$  via (3.28) of [7], where  $s_k = \|\mathbf{w}_{k+1} - \mathbf{w}_k\| = \beta_k \lambda_k / \|\mathbf{d}_k\|$ ). Since on ELM LASSO  $f^* > -\infty$ , the first branch is excluded and we need the second. [7, Lemma 3.8] provides a concrete update strategy for  $(f_{\text{ref}}^k, \delta_k)$  — the sufficient-descent test on  $f_{k+1} \leq f_{\text{ref}}^k - \delta_k/2$ , the target-infeasibility test on  $r_k > R$ , and the accumulate- $r$  counter — which attains both  $\delta_k \rightarrow 0$  and  $\sum_k s_k = \infty$ . Lines 16–21 of Algorithm 2 implement that strategy verbatim under the identification  $\mu \leftrightarrow \rho$ , so condition (3.26) holds.

*Subgradients uniformly bounded.* The convex objective  $f$  of ELM LASSO is coercive and continuous; the iterates  $\{\mathbf{w}_i\}$  remain in a compact sublevel set  $S_{c_0}$  for some  $c_0 \geq f(\mathbf{w}_0)$  (verification below), and  $\partial f$  is uniformly bounded on  $S_{c_0}$  by Lipschitz continuity of convex functions on compact sets [11, Slide 9].

Conditions (2.13), (3.5), (3.26) being verified with  $\sigma^* = 0$ , [7, Theorem 3.7] gives  $f_{\text{ref}}^\infty \leq f^* + \sigma^* = f^*$ , so  $\bar{f}_i \rightarrow f^*$ . The asymptotic rate (3.8) is then the classical Polyak rate [4, Thm. 3.2], [10, Slide 13] applied to the regime where the target-level estimate has sharpened to  $f^*$  (Lemma 3.8 of [7] guarantees  $\delta_k \rightarrow 0$ , so eventually the step reduces to the standard Polyak step with  $f^*$ ); the factor  $1/\gamma_{\min}$  is the explicit price of the deflection floor, which contracts the effective stepsize by  $\beta_i = \min(1, \gamma_i) \geq \gamma_{\min}$  at every iteration. On ELM LASSO with  $\gamma_{\min} = 0.05$  this inflates the iteration count to reach a given  $\varepsilon$  by 400× relative to the unfloored Polyak rate, but the  $O(L^2/\varepsilon^2)$  order is preserved.  $\square$

## Verification of theorem assumptions for ELM LASSO.

1. **Convexity.**  $\nabla^2 g = \mathbf{X}^T \mathbf{X} \succeq 0$ , the  $\ell_1$  norm is convex, so  $f$  is convex [12, Slides 21–24]. Under the standard ELM assumption  $M \geq H$  with  $\mathbf{X}$  full column rank [15],  $\nabla^2 g \succ 0$  and  $f$  is  $\mu$ -strongly convex with  $\mu = \lambda_{\min}(\mathbf{X}^T \mathbf{X}) > 0$  [11, Slide 12], [12, Slides 21–24]. SGPTL exploits neither acceleration nor proximal structure, so its guarantee remains  $O(L^2/\varepsilon^2)$ .
2. **Bounded subgradients.** The  $\ell_1$  part is bounded,  $\|\lambda_{\text{LASSO}} \mathbf{s}\|_2 \leq \lambda_{\text{LASSO}} \sqrt{H}$ . The smooth part  $\mathbf{X}^T(\mathbf{X}\mathbf{w} - \mathbf{y})$  grows with  $\|\mathbf{w}\|$ , so  $f$  is not globally Lipschitz. But  $f$  is coercive, every sublevel set is compact, and the target-level mechanism keeps  $f(\mathbf{w}_i) - (f_{\text{ref}} - \delta)$  on the same scale as  $f(\mathbf{w}_i) - f^*$ , so the iterates stay in an enlarged sublevel set  $S_{c_0}$  with  $c_0 \geq f(\mathbf{w}_0)$ . Lipschitz continuity of  $f$  on  $S_{c_0}$  [11, Slide 9] gives the uniform bound  $L$ .

**Complexity.** The lower bound for first-order methods on nonsmooth convex problems is  $\Omega(L^2/\varepsilon^2)$  [10, Slide 8], [4]; Algorithm 2 attains it. Deflection improves constants, not the order.

### 3.6 Expected practical behavior on the ELM LASSO problem

SGPTL is non-monotone: only  $\bar{f}_i = \min_{h \leq i} f(\mathbf{w}_h)$  decreases to  $f^*$ , so semilog plots use  $\log(\bar{f}_i - f^*)$  and the returned iterate is  $\mathbf{w}_{\text{best}}$ . The rate is sublinear  $O(L^2/\varepsilon^2)$ , so accuracies below  $10^{-6}$  are unrealistic. The per-iteration cost is  $O(MH)$  (two matvecs with  $\mathbf{X}$ ) against  $O(H^3)$  for IRLS’s Cholesky solve; DSM is CPU-competitive only at large  $H$  and moderate accuracy.

No reliable subgradient stopping test exists:  $0 \in \partial f(\mathbf{w}^*)$  does not force  $\|\mathbf{g}_i\| \rightarrow 0$ , so we stop on  $i_{\text{max}}$  or on a flat-window test on  $\bar{f}_i$ . Neither method produces exact zeros: IRLS pins them at the safety floor  $\varepsilon_{\text{thr}} = 10^{-8}$  in (2.5), SGPTL leaves them at the residual-rate level  $\sim L/\sqrt{i}$  ( $10^{-3}$ – $10^{-4}$  at our budgets). The fair comparison uses the same post-termination threshold on both (Section 5.4). Larger  $\lambda_{\text{LASSO}}$  amplifies oscillations through  $\mathbf{s}$  and inflates the iteration count.

## Chapter 4

# Algorithm Comparison

Both IRLS (Chapter 2) and the Deflected Subgradient Method (Chapter 3) solve the same LASSO problem

$$\min_{\mathbf{w} \in \mathbb{R}^H} f(\mathbf{w}) = \frac{1}{2} \|\mathbf{X}\mathbf{w} - \mathbf{y}\|_2^2 + \lambda_{\text{LASSO}} \|\mathbf{w}\|_1, \quad (4.1)$$

but handle the non-smoothness of the  $\ell_1$  penalty in opposite ways.

### 4.1 Core strategy: how each algorithm handles non-smoothness

#### IRLS — smoothing via majorization.

IRLS replaces  $f$  with a smooth quadratic majorant  $Q(\mathbf{w}, \mathbf{w}_k)$  tight at  $\mathbf{w}_k$  (Section 2.2), reducing each step to a weighted ridge problem solved in closed form via the normal equations (Section 2.3). The  $\ell_1$  penalty is absorbed into diagonal weights  $(\mathbf{W}_k)_{ii}$  that blow up where  $|w_{k,i}|$  is small, driving those components to zero (Section 2.1).

#### DSM — direct subgradient with memory.

DSM descends along a subgradient  $\mathbf{g} \in \partial f(\mathbf{w}_i)$  (Section 3.5.1), deflected with the previous direction (Section 3.1.1) to damp zig-zag, with a Polyak step using a self-adapting target estimate of  $f^*$  (Section 3.3).

IRLS solves a sequence of smooth problems *exactly* and is monotone in  $f$ ; DSM operates on  $f$  directly, accepts  $f(\mathbf{w}_i)$  may increase, and tracks progress via the record value  $\bar{f}^i$ .



## 4.2 Convergence behaviour

### 4.2.1 Monotonicity

MM guarantees IRLS is **monotone**:  $f(\mathbf{w}_{k+1}) \leq f(\mathbf{w}_k)$  (Section 2.2); a non-decreasing plot signals a bug, and an iterate-change criterion reliably stops the algorithm. DSM is **non-monotone**: only the record  $\bar{f}^i = \min_{h \leq i} f(\mathbf{w}_h)$  is monotone, so DSM returns  $\mathbf{w}_{\text{best}}$  and stops on a fixed budget.

### 4.2.2 Rate

IRLS achieves a **linear** rate (Section 2.5); DSM achieves the optimal **sublinear**  $O(1/\varepsilon^2)$  rate for nonsmooth convex functions (Theorem 3.1). Tightening  $\varepsilon$  by one decade costs  $\sim 100\times$  more DSM iterations versus a constant for IRLS.

### 4.2.3 Convergence on the ELM problem

For the ELM problem  $\mathbf{X} = \sigma(\mathbf{X}_{\text{origin}} \mathbf{W}_1^T)$  has full column rank (standard ELM condition when  $M \geq H$  with random  $\mathbf{W}_1$  and nonlinear  $\sigma$ ), so  $f$  is  $\mu$ -strongly convex with unique minimizer  $\mathbf{w}^*$ . IRLS iterates  $\mathbf{w}_k \rightarrow \mathbf{w}^*$  at a linear rate governed by the weighted normal-equation conditioning (Theorem 2.2); Theorem 3.1 only guarantees  $\bar{f}_i \rightarrow f^*$ , with iterates that may oscillate. Strong convexity would allow  $O(L^2/(\mu\varepsilon))$  [11, Slide 12], but SGPTL does not exploit it (Section 3.5.2).

## 4.3 Computational cost

### 4.3.1 Per-iteration cost

| Operation                  | IRLS                                  | DSM     |
|----------------------------|---------------------------------------|---------|
| Weight / direction update  | $O(H)$                                | $O(H)$  |
| Matrix-vector product(s)   | —                                     | $O(MH)$ |
| Linear system solve        | $O(H^3)$ (Cholesky) or $O(pH^2)$ (CG) | —       |
| <b>Total per iteration</b> | $O(H^3)$                              | $O(MH)$ |

Table 4.1: Per-iteration computational cost of IRLS and DSM.

IRLS solves an  $H \times H$  SPD system per iteration (Section 2.3):  $O(H^3/3)$  via Cholesky, or  $O(pH^2)$  via  $p$  CG steps when  $H$  is large and  $\mathbf{Q}_k$  well-conditioned. The matrices  $\mathbf{A} = \mathbf{X}^T \mathbf{X}$  and  $\mathbf{b} = \mathbf{X}^T \mathbf{y}$  are precomputed once at  $O(MH^2)$  and  $O(MH)$ .

DSM uses two matrix-vector products per iteration ( $\mathbf{X}\mathbf{w}_i$  and  $\mathbf{X}^T(\cdot)$ , each  $O(MH)$ ), no precomputation (Section 3.5.1).

### 4.3.2 Total cost

With iteration counts  $k_{\text{IRLS}}$ ,  $k_{\text{DSM}}$ :

$$\begin{aligned}\text{IRLS total: } & O(MH^2) + k_{\text{IRLS}} \cdot O(H^3), \\ \text{DSM total: } & k_{\text{DSM}} \cdot O(MH).\end{aligned}$$

Linear vs sublinear gives  $k_{\text{IRLS}} \ll k_{\text{DSM}}$ , so DSM is competitive only when  $H^2 \gg M$  and the Cholesky cost dominates.

On the ELM problem  $M \gg H$  is typical, so  $O(MH^2)$  precomputation dominates IRLS's total. Total costs become  $O(MH^2)$  vs  $O(k_{\text{DSM}} \cdot MH)$ , and IRLS wins whenever  $k_{\text{DSM}} \gtrsim H$ . Since Theorem 3.1 forces  $k_{\text{DSM}} = O(L^2/\varepsilon^2)$  with  $L$  growing in  $H$  (the  $\ell_1$  subgradient has norm  $\leq \lambda_{\text{LASSO}}\sqrt{H}$ ), this inequality holds at every  $H$  we test for any reasonable accuracy.

## 4.4 Solution quality

### 4.4.1 Sparsity

IRLS yields numerically sparse iterates as a byproduct of its weighted least-squares structure: small components accumulate large weights and are driven to exactly zero (within  $\varepsilon_{\text{thr}}$ ) at convergence, no post-processing (Section 2.7). DSM gives only **approximate sparsity**; an explicit thresholding step is needed (Section 3.6).

### 4.4.2 Accuracy

At equal budget IRLS reaches much higher accuracy on small-to-moderate problems thanks to its linear rate. DSM is competitive only when  $H^2 \gg M$  (Table 4.1).

## 4.5 Comparison summary for the ELM problem

On the project's regime ( $M \gg H$ ,  $\mathbf{W}_1$  fixed, offline training) IRLS is preferred: the  $O(MH^2)$  precomputation of  $\mathbf{A}$  amortizes over a few cheap Cholesky solves, sparsity comes for free, the single knob is  $\varepsilon_{\text{thr}}$ , and monotone  $f$  doubles as a correctness check. DSM is the better choice only when  $H^2 \gg M$  or  $\mathbf{X}^T\mathbf{X}$  does not fit in memory, and the accuracy target is moderate ( $\varepsilon \sim 10^{-3}$ ).

| <b>Criterion</b>   | <b>IRLS</b>                | <b>DSM</b>                     |
|--------------------|----------------------------|--------------------------------|
| Approach           | Smoothing (MM)             | Direct subgradient             |
| Monotone           | Yes                        | No (record value only)         |
| Convergence rate   | Linear                     | Sublinear $O(1/\varepsilon^2)$ |
| Per-iteration cost | $O(H^3)$                   | $O(MH)$                        |
| Precomputation     | $O(MH^2)$                  | None                           |
| Exact sparsity     | Yes                        | No (needs thresholding)        |
| Parameters to tune | $\varepsilon_{\text{thr}}$ | $\delta_0, \rho, R, \beta$     |
| Stopping criterion | Reliable (iterate change)  | Unreliable (fixed budget)      |
| Preferred when     | High accuracy, $H \ll M$   | Large $H$ , moderate accuracy  |

Table 4.2: Summary comparison of IRLS and DSM on the ELM LASSO problem.

## Chapter 5

# Experimental Results

This chapter reports experiments on the IRLS and SGPTL implementations of Chapters 2 and 3, against the predictions of Chapter 4: linear rate with natural sparsity for IRLS, sublinear non-monotone progress with approximate sparsity for SGPTL, and the  $O(MH^2) + k_{\text{IRLS}} \cdot O(H^3)$  versus  $k_{\text{SGPTL}} \cdot O(MH)$  cost trade-off.

### 5.1 Experimental setup

**Implementation.** The codebase under `progetto/code/src/` is organised in five modules: `lasso_utils.py` (shared objective, smooth-part gradient, soft-threshold subgradient selection), `linear_solvers.py` (Cholesky via `scipy.linalg.cho_factor/cho_solve` and a hand-rolled Jacobi-preconditioned CG), `irls.py` (Algorithm A1 driver), `deflected_subgradient.py` (Algorithm A2 with deflection floor  $\gamma_{\min}$ , clip  $\beta_i = \min(\beta, \gamma_i)$ , Polyak target-level), and `elm.py` (random hidden layer plus activations). Core algorithms are implemented from scratch; only NumPy/SciPy primitives are reused. A pytest suite under `progetto/code/tests/` cross-checks both algorithms against `sklearn.linear_model.Lasso` and verifies surrogate-descent and record-monotonicity invariants iterate-by-iterate.

**Reference solution.** We compute the reference  $\mathbf{w}^*$  via `sklearn.linear_model.Lasso` (coordinate descent, `tol=10-12`, `105` iterations, `fit_intercept=False`). Sklearn minimises

$$f_{\text{sklearn}}(\mathbf{w}) = \frac{1}{2M} \|\mathbf{X}\mathbf{w} - \mathbf{y}\|_2^2 + \alpha_{\text{sklearn}} \|\mathbf{w}\|_1, \quad (5.1)$$

which shares an arg-min with our  $f$  in eq. (1.3) when  $\alpha_{\text{sklearn}} = \lambda_{\text{LASSO}}/M$ . We then evaluate  $f^* = f(\mathbf{w}^*)$  directly from eq. (1.3). The returned  $\mathbf{w}^*$  satisfies the KKT condition (1.4) to below  $10^{-3}$  on every instance.

#### 5.1.1 Datasets

The experiments use three distinct instances.

*Synthetic LASSO benchmark.* Generated by `src/data_generation.py`. We sample a sparse  $\mathbf{w}_{\text{true}} \in \mathbb{R}^H$  (10–15% non-zero, entries  $\sim \mathcal{N}(0, 1)$ ), build a random  $\mathbf{X} \in \mathbb{R}^{M \times H}$  with i.i.d. Gaussian columns normalised to unit  $\ell_2$  norm, and set  $\mathbf{y} = \mathbf{X}\mathbf{w}_{\text{true}} + \boldsymbol{\varepsilon}$  with  $\boldsymbol{\varepsilon} \sim \mathcal{N}(0, 0.05^2 \mathbf{I})$ . This is the *best-case scenario* for LASSO: column-normalised Gaussian designs are mutually incoherent [14] and keep  $\kappa(\mathbf{X}^\top \mathbf{X})$  moderate when  $M \geq 2H$  (e.g.  $\kappa \approx 14$  at  $H = 100$ ,  $M = 300$  from Marchenko–Pastur). The ground-truth sparsity makes support recovery clean, the noise level  $\sigma = 0.05$  puts the SNR in the 5–10% range across tested sizes. The benchmark is well-conditioned and validates the algorithms in a benign regime; it does *not* reproduce the conditioning of ELM-transformed real data, where the sigmoid activation typically inflates  $\kappa$  by orders of magnitude.

*ELM-transformed real datasets (diabetes, california).* We pull `diabetes` ( $n = 442$ ,  $d = 10$ ) [9] and `california_housing` ( $n = 20640$ ,  $d = 8$ ) [17] from scikit-learn. Each set is split 80/20 into train/test, both features and target standardised on the train split, and the ELM transformation  $\mathbf{X}_{\text{hidden}} = \sigma(\mathbf{X} \mathbf{W}_1^\top)$  with  $H$  sigmoid units and a fixed random  $\mathbf{W}_1$  produces the LASSO design matrix. We use  $H = 200$  for the main experiments and  $H = 50$  for the  $\delta_0$  family sweep of Section 5.3, where a smaller matrix lets us cross-validate  $f^*$  with an independent solver (`cvxpy`.CLARABEL interior point); the cross-validation agrees with the IRLS-converged value to relative error below  $10^{-9}$  on `diabetes`  $H = 50$  (and below  $10^{-7}$  on the synthetic instances of  $H = 100$ ,  $M = 300$  where it is also feasible). The sigmoid breaks the unit-norm column property and inflates  $\kappa(\mathbf{X}_{\text{hidden}}^\top \mathbf{X}_{\text{hidden}})$  to  $\sim 10^6$  on both datasets, so the ELM matrices are markedly worse-conditioned than the synthetic benchmark.

*Limitations of the benchmark.* Three to flag explicitly. (i) The synthetic benchmark is essentially the easy regime: no multicollinearity, no heavy-tailed features, deliberately well-conditioned. Results that hold there are not automatic on harder instances, and we test them on the real ELM matrices when feasible. (ii) The ELM-transformed real datasets have no ground-truth sparsity, so the sparsity claims of Section 5.4 apply only to the synthetic instance; on the ELM real data we report training-objective gaps and test MSE. (iii) On `california` at  $\lambda_{\text{LASSO}} = 0.1$  the OLS warm start already coincides with  $f^*$  to within  $5 \cdot 10^{-1}$  on the ELM matrix (the  $\ell_1$  term is weak relative to the data scale on this dataset), so warm-started SGPTL has nothing to optimise and the configuration is uninformative for sensitivity studies. For the  $\delta_0$  sweep of Section 5.3 we therefore run `california` from cold start  $\mathbf{w}_0 = \mathbf{0}$ , where the initial gap is  $\sim 8 \cdot 10^3$  and SGPTL has substantive work to do; we cross-validated this choice by showing that SGPTL reduces the cold-start gap by 99%+ on this matrix at every  $\lambda \in \{10^{-3}, 10^{-2}, 0.1, 1, 10\}$  tested. The `california` comparison study itself is in Section 5.7 and uses OLS warm start for consistency with IRLS.

**Timing protocol.** Wall-clock times are taken with `time.perf_counter` and exclude the data-generation step. They reflect single-thread Python execution without explicit BLAS-thread tuning.

**The deflection floor in practice.** Section 3.2 motivates the lower bound  $\gamma_i \geq \gamma_{\min}$  as a structural ingredient for the clip  $\beta_i = \min(\beta, \gamma_i)$ . We test the prediction empirically across a  $3 \times 5 \times 2$  grid: three sizes  $(H, M) \in \{(50, 150), (100, 300), (200, 600)\}$ , five seeds, two configurations  $\gamma_{\min} \in \{0, 0.05\}$  (`experiments/experiment_gamma_floor.py`). The pattern is uniform across the 30 runs. With  $\gamma_{\min} = 0$  the greedy (3.5) drives  $\gamma_i$  below 0.01 for  $\sim 99\%$  of iterations;  $\beta_i = \min(1, \gamma_i)$  collapses with it,  $f(\mathbf{w}_i) - (f_{\text{ref}} - \delta)$  becomes non-positive, the numerator-skip branch fires for  $\sim 98\%$  of iterations,  $\delta$  contracts only once or twice over 4000 steps, and the record gap stalls in  $[2 \cdot 10^{-2}, 1 \cdot 10^{-1}]$ . With  $\gamma_{\min} = 0.05$  the floor binds where the greedy would otherwise hit zero, the numerator-skip branch never fires,  $\delta$  contracts  $\sim 19$  times over the same budget, and the gap descends to  $[6 \cdot 10^{-5}, 3 \cdot 10^{-4}]$ . The 20–50 $\times$  gap separation and the  $\sim 99\%$  vs 0% splits in skip-fire rate and  $\gamma_i \leq 0.01$  rate hold uniformly across sizes and seeds, so the failure mode is structural to ELM LASSO with the greedy  $\gamma$ , not an artefact of one instance. Figure 5.1 shows the  $\gamma_i$  histogram, the  $\delta$  staircase and the gap trajectory for the medium size.

The calibration of  $\gamma_{\min}$  within the floored range is the secondary question. A sweep  $\gamma_{\min} \in \{0.05, 0.1, 0.2, 0.5\}$  keeps the final gap in  $10^{-7}$ – $10^{-5}$ , with  $\gamma_{\min} = 0.05$  marginally best (Figure 5.2).

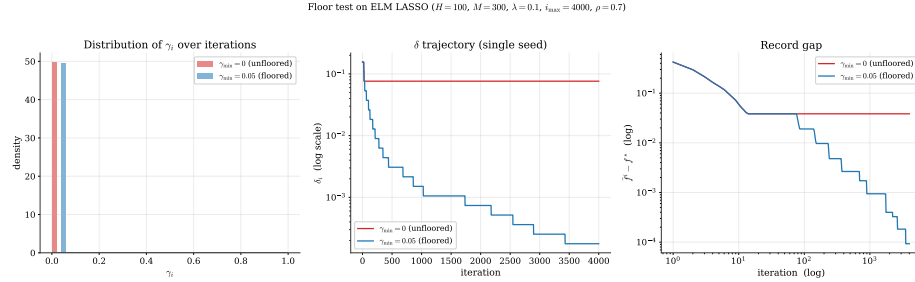


Figure 5.1: Floor test on ELM LASSO ( $H = 100$ ,  $M = 300$ ,  $\lambda_{\text{LASSO}} = 0.1$ ,  $i_{\max} = 4000$ ,  $\rho = 0.7$ ). Left:  $\gamma_i$  distribution over iterations — the unfloored run piles  $\sim 99\%$  of its mass at  $\gamma_i \approx 0$ , the floored run lives in  $[\gamma_{\min}, 1]$ . Centre:  $\delta$  over iterations — the unfloored  $\delta$  contracts twice and then drifts as the skip branch keeps firing, the floored  $\delta$  descends in the staircase the theory expects. Right: record gap — two orders of magnitude separation, stable across the 15 seeds tested per configuration.

### 5.1.2 Implementation note: from the first submission to the theory-pure rule

The first submission carried, on top of Algorithm 2, an iteration-count  $\delta$ -contraction fallback, an overshoot rescue snapping  $\mathbf{w}$  back to  $\mathbf{w}_{\text{best}}$ , and the default  $R = 10\sqrt{i_{\max}} \approx 894$  at  $i_{\max} = 8000$  — far above any realistic per-iteration step length ( $\alpha_i \|\mathbf{d}_i\| \sim 10^{-3}$ ), so the  $r > R$  branch never fired and the iteration-count fallback drove every  $\delta$ -contraction. The instructor flagged both safeguards as

inconsistent with [7, Lemma 3.8], which requires every contraction to follow  $r_k > R$  travel. This submission removes both workarounds and sets  $R = 1$  to match the natural step length; on the convergence instance this gives  $\sim 22$  contractions over 8000 iterations at  $\rho = 0.7$  (against  $\sim 95$  under the old fallback and 0 under the old  $R$ ).  $R = 1$  is the one calibration outside the pseudocode; the theory requires only  $R > 0$ . The new gap on the convergence instance is  $4.8 \cdot 10^{-5}$  against the first submission's  $\sim 6.6 \cdot 10^{-8}$ ; the previous figure was a numerical artefact of the iteration-count fallback. The qualitative comparison (IRLS faster, SGPTL cheaper per iteration) is preserved. First-submission artefacts are retained under `progetto/code/results_old_submission/`.

**Warm vs cold start for SGPTL.** Under the old  $R$  cold start dominated (warm start triggered blow-ups); under  $R = 1$  the two are essentially neutral. Figure 5.2 shows both starts on the convergence instance: the cold start ends at  $2.4 \cdot 10^{-5}$  with 26  $\delta$ -contractions, the warm start at  $4.8 \cdot 10^{-5}$  with 22. Across similar instances neither is uniformly better. We keep the OLS warm start as the default for SGPTL because it is the same starting point used by IRLS, which makes the comparison apples-to-apples.

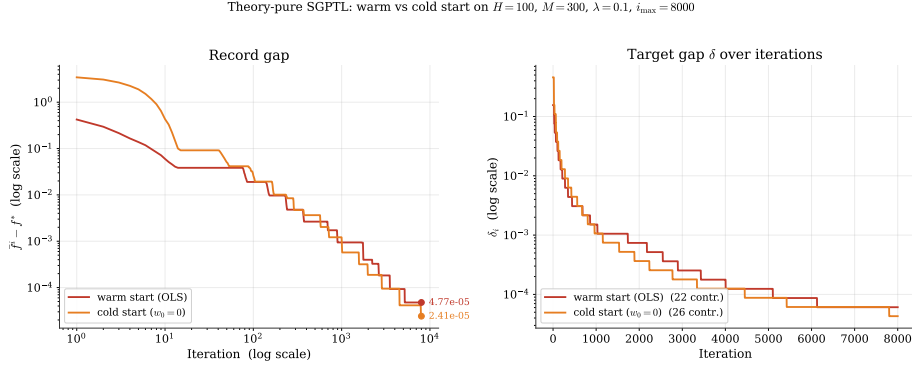


Figure 5.2: Theory-pure SGPTL with the OLS warm start versus cold start ( $\mathbf{w}_0 = \mathbf{0}$ ) on the same synthetic instance ( $H = 100$ ,  $M = 300$ ,  $\lambda_{\text{LASSO}} = 0.1$ ,  $i_{\max} = 8000$ ,  $\rho = 0.7$ ,  $\gamma_{\min} = 0.05$ ,  $R = 1$ ). Left panel: record gap  $\hat{f}^i - f^*$  on log-log axes; both starts produce sublinear descent. Right panel:  $\delta_i$  staircase (26 contractions cold, 22 warm). Final gaps within a factor of two; warm start is the default for consistency with IRLS.

## 5.2 Convergence behaviour

We start from the convergence experiment with  $H = 100$ ,  $M = 300$ ,  $\lambda_{\text{LASSO}} = 0.1$ , sparsity 10%. Both algorithms use the OLS warm start  $\mathbf{w}_0 = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$  (Section 3.4).

**Cost of the warm start.** The OLS warm start solves  $\mathbf{X}^\top \mathbf{X} \mathbf{w}_0 = \mathbf{X}^\top \mathbf{y}$  via Cholesky at cost  $O(MH^2) + O(H^3/3)$  — one IRLS-iteration of work on the convergence instance, at most  $\sim 40\%$  of the SGPTL budget on the largest scalability instance, well below 1% elsewhere. Both algorithms pay it once and we include it in their wall-clock totals. On the convergence instance  $f(\mathbf{0}) \approx 76$ ,  $f(\mathbf{w}_0) \approx 1.56$ ,  $f^* \approx 1.13$ : the warm start lands  $\sim 180\times$  closer to the optimum than  $\mathbf{0}$ , a gap that SGPTL would need thousands of iterations to recover under the  $O(L^2/\varepsilon^2)$  rate. Using the same  $\mathbf{w}_0$  for both algorithms also removes start as a confounder in the comparison.

Figure 5.3 shows the gap  $f(\mathbf{w}_k) - f^*$  versus iteration. IRLS produces a straight line on the semilog left panel — the linear MM rate of Section 2.5 — reaching  $1.5 \cdot 10^{-6}$  in 100 iterations, with iterate change  $5.6 \cdot 10^{-6}$ .

SGPTL (8000 iterations,  $\beta = 1$ ,  $\gamma_{\min} = 0.05$ ,  $\delta_0 = 0.1 f(\mathbf{w}_0)$ ,  $\rho = 0.7$ ) gives the right panel: the current gap oscillates as expected, the record descends along the sublinear envelope to a final  $4.8 \cdot 10^{-5}$ . Each additional decade multiplies the iteration count by  $\sim 100$ , so the gap floor is budget-limited.

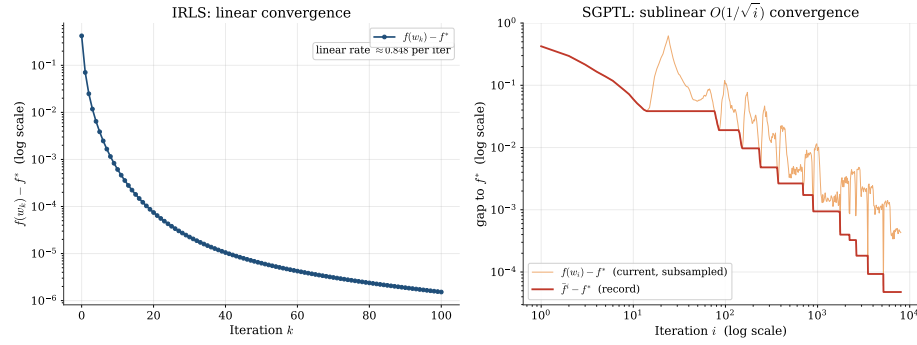


Figure 5.3: Optimality gap versus iteration count on a problem with  $H = 100$ ,  $M = 300$ ,  $\lambda_{\text{LASSO}} = 0.1$ . Left: IRLS on a semilog axis; the in-panel rate annotation is the empirical per-iteration multiplicative reduction sampled in the linear region. Right: SGPTL on log–log axes, with the current value  $f(\mathbf{w}_i) - f^*$  (orange, oscillating; subsampled log-uniformly so the late-iteration density does not collapse into a solid band) overlaid on the record  $\bar{f}^i - f^*$  (red). The record traces the sublinear  $O(1/\sqrt{i})$  envelope predicted by Theorem 3.1: gaining each decade of accuracy multiplies the iteration count by  $\sim 100$ .

Figure 5.4 replots against wall-clock time. On this size SGPTL’s per-iteration advantage ( $O(MH)$  vs  $O(H^3)$ ) does not compensate for the slower rate: IRLS reaches  $10^{-6}$  in  $\sim 3$  ms, SGPTL falls below  $10^{-2}$  within  $\sim 10$  ms and then converges sublinearly to  $\sim 5 \cdot 10^{-5}$  over the remaining  $\sim 150$  ms.



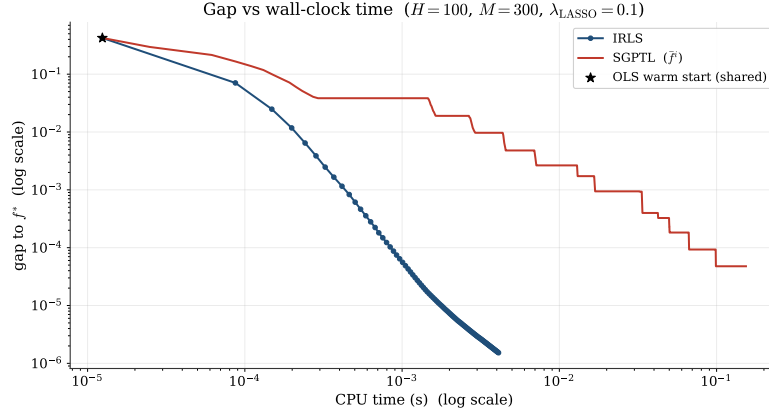


Figure 5.4: Optimality gap versus CPU time on the same instance as Figure 5.3, log-log axes.

Non-monotonicity (Section 3.3) is visible empirically in Figure 5.5.

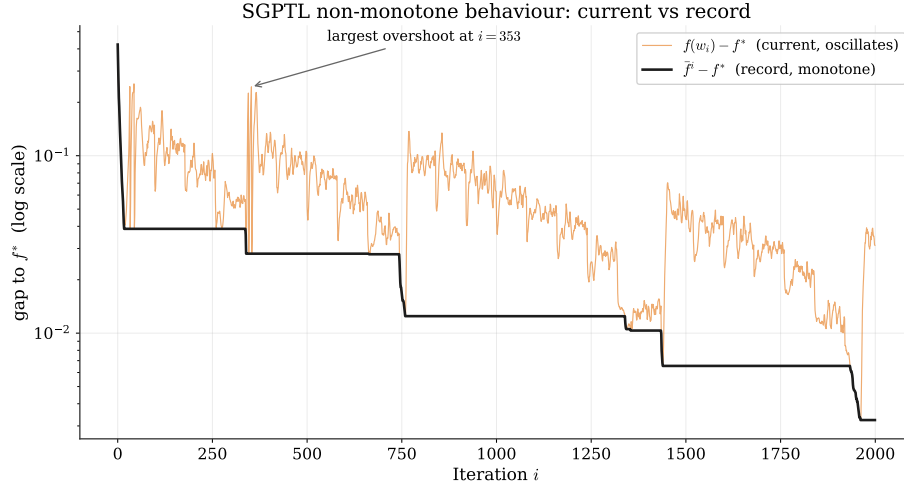


Figure 5.5: SGPTL on the convergence instance, first 2000 iterations, semilog  $y$  axis. The current gap  $f(\mathbf{w}_i) - f^*$  (orange) spikes well above the record gap  $\tilde{f}^i - f^*$  (black) at several points throughout the run, while the record is monotone non-increasing by construction. The largest overshoot is annotated. This is the practical reason why the algorithm returns  $\mathbf{w}_{\text{best}}$  rather than the last iterate.

## 5.3 Parameter sensitivity

### 5.3.1 IRLS: $\varepsilon_{\text{thr}}$ and $\lambda_{\text{LASSO}}$

The two parameters acting on IRLS are the safety threshold  $\varepsilon_{\text{thr}}$  in (2.5) and  $\lambda_{\text{LASSO}}$ . We sweep both at  $H = 100$ ,  $M = 400$ , 100 iterations.

Figure 5.6 (left): for  $\varepsilon_{\text{thr}} \in \{10^{-4}, 10^{-6}, 10^{-8}, 10^{-10}, 10^{-12}\}$  the final gap tracks  $\varepsilon_{\text{thr}}$  ( $1.4 \cdot 10^{-6}$ ,  $1.5 \cdot 10^{-8}$ ,  $5.3 \cdot 10^{-10}$  for  $10^{-6}, 10^{-8}, 10^{-10}$ );  $10^{-4}$  is too loose ( $1.4 \cdot 10^{-4}$ , no sparsity);  $10^{-12}$  hits floating-point limits. Sparsity stabilises at 86% (reference 88%). The default  $10^{-8}$ – $10^{-10}$  used elsewhere is the sweet spot.

Right:  $\lambda_{\text{LASSO}} \in \{0.01, 0.05, 0.1, 0.5, 1.0\}$ . Convergence speed is roughly invariant ( $\lambda$  enters  $\mathbf{Q}_k$  as a bounded conditioning factor); sparsity moves from 11% to 95%, monotone in  $\lambda$  as predicted by (1.4).

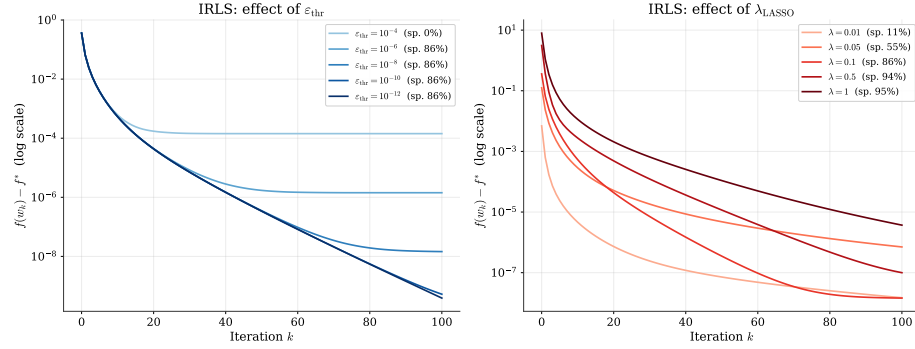


Figure 5.6: IRLS sensitivity. Left: effect of  $\varepsilon_{\text{thr}}$  on convergence and sparsity at  $\lambda_{\text{LASSO}} = 0.1$ . Right: effect of  $\lambda_{\text{LASSO}}$  at  $\varepsilon_{\text{thr}} = 10^{-8}$ .

### 5.3.2 IRLS: linear solver choice (Cholesky vs. Conjugate Gradient)

Chapter 2 identifies Cholesky ( $O(H^3/3)$ ) and CG ( $O(pH^2)$  per CG step) as candidate inner solvers for  $\mathbf{Q}_k \mathbf{w}_{k+1} = \mathbf{b}$ ; the trade-off depends on  $H$  and  $\kappa(\mathbf{Q}_k)$ . We compare them on the instance  $H = 100$ ,  $M = 400$ ,  $\lambda_{\text{LASSO}} = 0.1$ , OLS warm start,  $\varepsilon_{\text{thr}} = 10^{-8}$ ,  $k_{\text{max}} = 300$  (Table 5.1).

| Solver   | Total time (ms) | Iterations | Sparsity at $10^{-6}$ | Converged |
|----------|-----------------|------------|-----------------------|-----------|
| Cholesky | 11.7            | 265        | 86%                   | Yes       |
| CG       | 24.4            | 265        | 86%                   | Yes       |

Table 5.1: IRLS solver comparison on  $H = 100$ ,  $M = 400$ ,  $\lambda_{\text{LASSO}} = 0.1$ ,  $\varepsilon_{\text{thr}} = 10^{-8}$ ,  $k_{\text{max}} = 300$ , OLS warm start. Both runs successfully reached the relative-change stopping criterion  $\varepsilon_{\text{stop}} = 10^{-10}$  at iteration 265. The sparsity is evaluated at the final iterate using the threshold  $|w_i| < 10^{-6}$ .

Figure 5.7 plots the gap versus iteration and time.

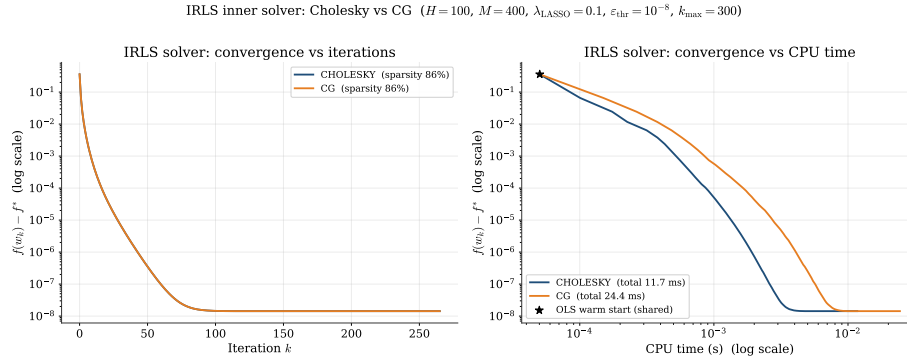


Figure 5.7: IRLS inner-solver comparison on  $H = 100$ ,  $M = 400$ ,  $\lambda_{\text{LASSO}} = 0.1$ ,  $\varepsilon_{\text{thr}} = 10^{-8}$ ,  $k_{\text{max}} = 300$ . Left: gap  $f(\mathbf{w}_k) - f^*$  versus iteration on a semilog  $y$  axis. Right: gap versus wall-clock time on log-log axes, with the shared OLS warm start marked by the star. Cholesky descends smoothly to  $\sim 10^{-8}$  ( $\sim \varepsilon_{\text{thr}}$ ) within  $\sim 11.7$  ms; CG achieves the exact same iterations trajectory but requires  $\sim 24.4$  ms.

**Analysis.** Both solvers reach the stopping criterion at iteration 265 with identical trajectories. Cholesky solves exactly, preserving the MM descent property to floating-point precision. CG is run with default `tol` =  $10^{-10}$ , `max_iter` =  $10H = 1000$ ; as IRLS drives inactive weights to the cap  $1/\varepsilon_{\text{thr}} = 10^8$ ,  $\kappa(\mathbf{Q}_k)$  blows up, so we apply a Jacobi preconditioner on the diagonal of  $\mathbf{Q}_k$  to neutralise the penalty-term scales. Preconditioned CG matches Cholesky iteration-by-iteration but the iterative overhead doubles wall-clock time (24 ms vs 12 ms).

**Recommendation.** In the ELM regime targeted here ( $M \gg H$ ,  $H \leq 1000$ ) Cholesky is the default: faster on the test instance, MM descent guaranteed unconditionally, no inner tolerance to tune. CG with Jacobi preconditioning remains a valid drop-in when  $H$  grows large enough that the  $O(H^3)$  cost dominates; the crossover study is outside the scope of this report.

### 5.3.3 SGPTL: $\delta_0$ and $\rho$

The two target-level parameters are  $\delta_0$  and  $\rho$  ( $\beta = 1$  fixed,  $R$  at default).

**Reference scale for  $\delta_0$ .**  $\delta_0$  must not depend on  $f^*$ . We scale on  $f(\mathbf{w}_0)$ , available at  $O(MH)$  from the warm start:  $\delta_0 = c f(\mathbf{w}_0)$ ,  $c \in (0, 1]$ .<sup>1</sup>

**Sensitivity sweeps.** Figure 5.9, left:  $c \in \{0.01, 0.05, 0.1, 0.5, 1.0\}$  at  $\rho = 0.7$  ( $H = 100$ ,  $M = 400$ , 5000 iterations) gives final gaps in  $[5.6 \cdot 10^{-5}, 1.6 \cdot 10^{-4}]$  — within one order of magnitude across two decades of  $\delta_0$ . Default  $c = 0.1$ .

Right:  $\rho \in \{0.3, 0.5, 0.7, 0.8, 0.9, 0.95\}$  at  $\delta_0 = 0.1 f(\mathbf{w}_0)$  yields gaps  $1.98 \cdot 10^{-5}$ ,  $2.28 \cdot 10^{-5}$ ,  $1.02 \cdot 10^{-4}$ ,  $1.51 \cdot 10^{-4}$ ,  $3.71 \cdot 10^{-4}$ ,  $8.12 \cdot 10^{-4}$  with 7, 12, 19, 30, 54, 91  $\delta$ -contractions. Small  $\rho$  wins here;  $\rho \geq 0.9$  leaves the target above  $f^*$  for most of the budget.

**Choice of default  $\rho$ .** Theorem 3.1 accepts any  $\rho \in (0, 1)$  with the same asymptotic  $O(L^2/\varepsilon^2)$  rate; the constant scales as  $1/\log(1/\rho)$ . The target-level literature [3, 13, 1], [10, Slide 14] recommends  $\rho \in [0.5, 0.7]$ . A smoke-test sweep across the synthetic instance and the two ELM matrices (**diabetes**, **california**) places the best  $\rho$  at 0.3, 0.3 and 0.7 respectively;  $\rho \in [0.3, 0.5]$  is within a factor of two of the best on every instance,  $\rho \geq 0.9$  loses by one to two orders of magnitude. We adopt  $\rho = 0.7$  as a robust default.

**Sensitivity of  $\delta_0$ : three families.** Since  $\delta_0 \propto f^*$  is inadmissible we compare three  $f^*$ -free choices against the oracle on five instances: the synthetic LASSO benchmark of Section 5.1.1 (five seeds), the ELM-transformed **diabetes** at  $H = 50$  and  $H = 200$ , and the ELM-transformed **california** at  $H = 50$  and  $H = 200$ . The families are defined relative to the actual starting point  $\mathbf{w}_0$  used by SGPTL on each instance: Family A,  $\delta_0 = c f_{\text{LASSO}}(\mathbf{w}_0)$  (default, admissible); Family B,  $\delta_0 = c \frac{1}{2} \|\mathbf{X}\mathbf{w}_0 - \mathbf{y}\|^2$  (admissible, smooth-part only); Family C,  $\delta_0 = c f^*$  (diagnostic oracle, not admissible as a tuning target). The sweep is  $c \in \{0.01, 0.05, 0.1, 0.5, 1\}$  on every instance. On four of the five instances  $f^*$  is CVXPY-verified (CLARABEL interior point, cross-validated against the IRLS-converged value with relative agreement below  $10^{-9}$ ); on **california**  $H = 200$  the size puts CVXPY beyond a reasonable budget and we rely on the IRLS-converged proxy.

*Starting point per instance.* The synthetic and **diabetes** runs use the OLS warm start, consistent with IRLS and with the rest of this chapter. On **california** at  $\lambda_{\text{LASSO}} = 0.1$  the OLS warm start already attains  $f^*$  to within  $5 \cdot 10^{-1}$  (the weak  $\ell_1$  regulariser leaves OLS near-optimal on this ELM matrix), so warm-started SGPTL has essentially nothing to optimise and the three families coincide trivially at the starting gap. To obtain a meaningful comparison we instead run **california** from the cold start  $\mathbf{w}_0 = \mathbf{0}$ , where the initial gap

<sup>1</sup>An earlier draft scaled on  $f^*$  ( $f(\mathbf{w}_0)$  exceeds  $f^*$  by  $\sim 20$ – $50\%$  on the test instances), which is inadmissible.

is  $\|\mathbf{y}\|^2/2 \approx 8.3 \cdot 10^3$  and SGPTL closes it by 99%+ over the iteration budget. The configuration per instance is summarised in Table 5.2.

*Empirical finding.* Across every instance, in the regime where SGPTL is in fact optimising, the Family A and Family C trajectories agree within a factor of two. On the synthetic instance (5 seeds, all  $c$ ) the ratio is in  $[0.50, 1.07]$ . On **diabetes**  $H = 50$  at  $c \leq 0.1$  (working regime, gap  $\sim 10^{-5}$ ) the ratio is in  $[0.73, 1.96]$ . On **diabetes**  $H = 200$  (pre-converged) it is in  $[0.87, 1.04]$ . On **california**  $H = 50$  cold the ratio is in  $[0.72, 1.41]$  and on **california**  $H = 200$  cold it is in  $[0.96, 1.07]$ . The admissible default  $\delta_0 = c f(\mathbf{w}_0)$  therefore matches the inadmissible oracle on every instance where the comparison is informative.

*The single uninformative regime.* On **diabetes**  $H = 50$  at  $c \geq 0.5$  all three families stall at a common plateau  $\bar{f}^i - f^* \approx 1.4 \cdot 10^{-1}$  (forcing the ratio to 1.000) because the over-aggressive  $\delta_0$  overshoots the sufficient-descent test from the first iteration and only the travel-distance counter drives progress; the three families behave identically simply because none of them recover from the initial overshoot. We report this as a regime limit of the  $\delta_0$ -sweep, not as a finding about A-vs-C calibration.

Family B coincides with Family A on **california** cold start (where  $\|\mathbf{w}_0\| = 0$  makes the  $\ell_1$  term vanish) and differs only mildly on the other instances. Family A with  $c = 0.1$  is the default.

| instance                    | seeds | $M \times H$       | start                            | $f^*$ source         |
|-----------------------------|-------|--------------------|----------------------------------|----------------------|
| synthetic                   | 5     | $300 \times 100$   | OLS warm                         | CVXPY-verified       |
| <b>diabetes</b> $H = 50$    | 1     | $354 \times 50$    | OLS warm                         | CVXPY-verified       |
| <b>diabetes</b> $H = 200$   | 1     | $354 \times 200$   | OLS warm                         | CVXPY-verified       |
| <b>california</b> $H = 50$  | 1     | $16512 \times 50$  | cold $\mathbf{w}_0 = \mathbf{0}$ | CVXPY-verified       |
| <b>california</b> $H = 200$ | 1     | $16512 \times 200$ | cold $\mathbf{w}_0 = \mathbf{0}$ | IRLS-converged proxy |

Table 5.2: Instances used in the  $\delta_0$  family sweep. The **california** runs use cold start because the OLS solution coincides with  $f^*$  to within machine-precision on this matrix at  $\lambda_{\text{LASSO}} = 0.1$ , so the warm-start configuration leaves SGPTL no gap to close. All runs share  $\lambda_{\text{LASSO}} = 0.1$ ,  $i_{\max} = 8000$ ,  $\rho = 0.7$ ,  $\gamma_{\min} = 0.05$ ,  $\beta = 1$ ,  $R = 1$ .

SGPTL:  $\delta_0$  family sweep ( $\rho = 0.7$ ,  $\gamma_{\min} = 0.05$ ,  $i_{\max} = 8000$ )

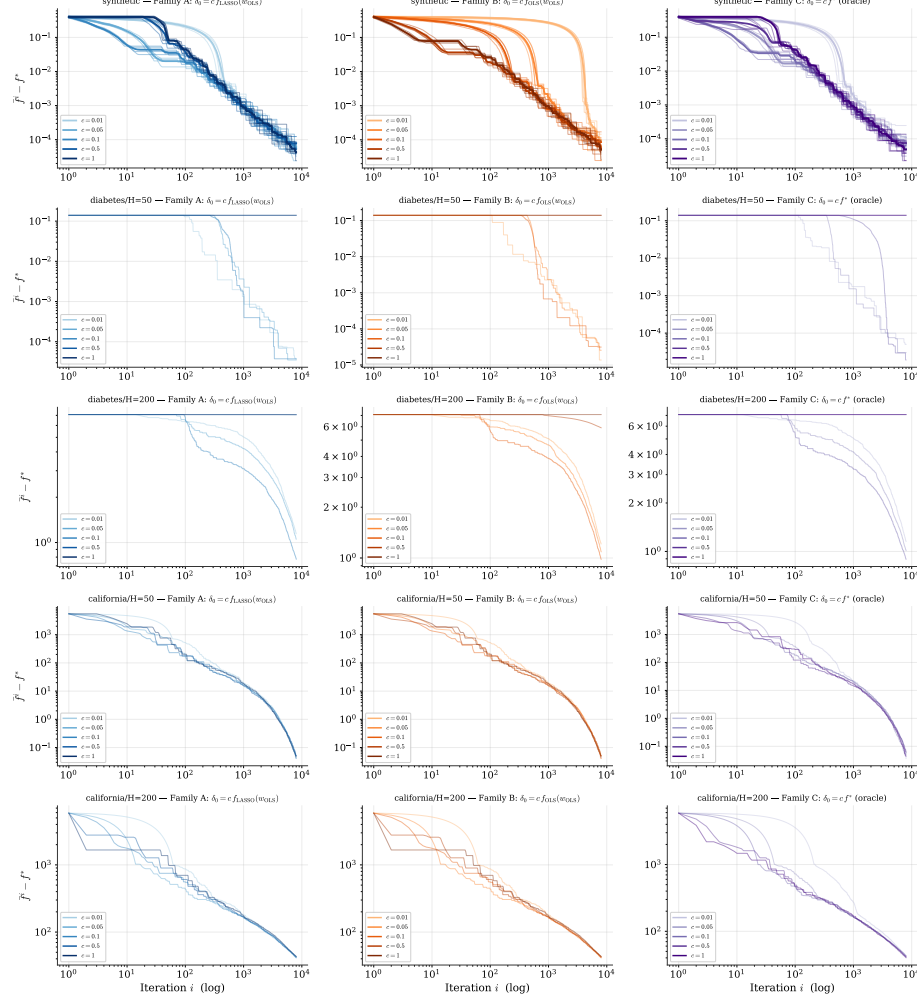


Figure 5.8: Sensitivity of the final record gap to the choice of  $\delta_0$ , three families across five instances (rows, top to bottom): **synthetic** LASSO (OLS warm start, 5 seeds with per-seed thin traces and aggregate-mean thick), **diabetes**  $H = 50$  (warm), **diabetes**  $H = 200$  (warm), **california**  $H = 50$  (cold start — see Table 5.2), **california**  $H = 200$  (cold). All runs use  $\lambda_{\text{LASSO}} = 0.1$ ,  $\rho = 0.7$ ,  $\gamma_{\min} = 0.05$ ,  $i_{\max} = 8000$ . Family A:  $\delta_0 = c f_{\text{LASSO}}(\mathbf{w}_0)$  (default, admissible). Family B:  $\delta_0 = c \frac{1}{2} \|\mathbf{X}\mathbf{w}_0 - \mathbf{y}\|^2$  (admissible). Family C:  $\delta_0 = c f^*$  (diagnostic oracle, not admissible). Across all rows the Family A vs Family C ratio falls in  $[0.5, 2.0]$  in every regime where SGPTL admits progress.

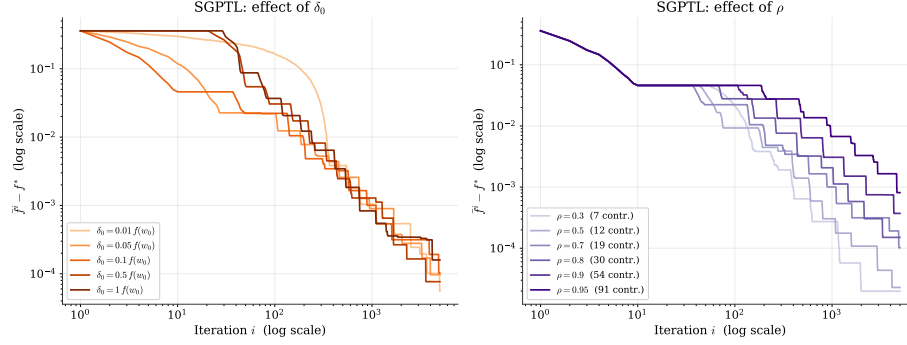


Figure 5.9: SGPTL sensitivity ( $H = 100$ ,  $M = 400$ , OLS warm start, 5000 iterations). Left: effect of  $\delta_0$  as a fraction of  $f(\mathbf{w}_0)$  at  $\rho = 0.7$  — final gaps in  $[5.6 \cdot 10^{-5}, 1.6 \cdot 10^{-4}]$  across two decades of  $\delta_0$ . Right: effect of  $\rho$  at  $\delta_0 = 0.1 f(\mathbf{w}_0)$ ; gaps and contraction counts grow monotonically with  $\rho$  (7 contractions at  $\rho = 0.3$ , 91 at  $\rho = 0.95$ ).  $\rho = 0.7$  chosen as default for robustness across the smoke-test grid (synthetic, diabetes, california).

## 5.4 Solution quality and sparsity

We compare solution quality on  $H = 50$ ,  $M = 200$ ,  $\lambda_{\text{LASSO}} = 0.1$ , reference sparsity 88%. IRLS reaches  $f - f^* \leq 10^{-6}$  in 101 iterations at  $\|\mathbf{w}_{\text{IRLS}} - \mathbf{w}^*\|_2 = 2.95 \cdot 10^{-6}$ ; SGPTL (warm start,  $\beta = 1$ ,  $\gamma_{\min} = 0.05$ ,  $\delta_0 = 0.1 f(\mathbf{w}_0)$ ,  $\rho = 0.7$ ) reaches  $1.0 \cdot 10^{-4}$ ,  $\sim 35\times$  looser.

**Sparsity measurement.** A  $|w_i| < 10^{-6}$  cutoff is unfair: IRLS pins inactive components at  $\sim \varepsilon_{\text{thr}} = 10^{-8}$ , SGPTL leaves them at  $\sim L/\sqrt{i} \in [10^{-4}, 10^{-3}]$  above the cutoff. We apply the same hard threshold to both methods and report precision/recall/F1 against the support of  $\mathbf{w}^*$  (Table 5.3).

| threshold | method       | sparsity | precision | recall | F1    |
|-----------|--------------|----------|-----------|--------|-------|
| $10^{-6}$ | IRLS         | 86%      | 0.857     | 1.000  | 0.923 |
| $10^{-6}$ | SGPTL        | 46%      | 0.222     | 1.000  | 0.364 |
| $10^{-6}$ | ground truth | 88%      | 1.000     | 1.000  | 1.000 |
| $10^{-3}$ | IRLS         | 88%      | 1.000     | 1.000  | 1.000 |
| $10^{-3}$ | SGPTL        | 88%      | 1.000     | 1.000  | 1.000 |
| $10^{-3}$ | ground truth | 88%      | 1.000     | 1.000  | 1.000 |

Table 5.3: Support recovery on the moderate problem ( $H = 50$ ,  $M = 200$ ,  $\lambda_{\text{LASSO}} = 0.1$ , reference sparsity 88%): same hard threshold applied to both algorithms, precision / recall / F1 against the support of  $\mathbf{w}^*$ . At  $10^{-3}$  both algorithms recover the reference support exactly. At  $10^{-6}$  IRLS leaves a handful of inactive components pinned around  $\varepsilon_{\text{thr}} = 10^{-8}$  but still in the  $10^{-7}$  band; SGPTL drops  $F_1$  to 0.364 because subgradient steps lack any pinning mechanism and the inactive coefficients sit at the residual-rate scale  $L/\sqrt{i} \sim 10^{-3}$ .

With the threshold matched to each method’s residual scale both algorithms recover the support exactly (the  $10^{-3}$  rows in Table 5.3); SGPTL requires the larger threshold to clear the  $L/\sqrt{i}$  floor (Section 3.6).

## 5.5 Scalability

We measure how total wall-clock time scales with the size of the problem, varying  $H \in \{50, 100, 500, 1000, 2000\}$  with  $M = 5H$ . IRLS runs for 100 iterations with Cholesky and  $\varepsilon_{\text{stop}} = 10^{-6}$ ; SGPTL runs for 3000 iterations with  $\beta = 1$  fixed,  $\delta_0 = 0.1 f(\mathbf{w}_0)$ ,  $\rho = 0.7$  (the revised defaults of Section 5.3). Table 5.4 collects the numbers.

| $H$  | $M$   | $t_{\text{IRLS}}$ (s) | gap <sub>IRLS</sub> | $t_{\text{SGPTL}}$ (s) | gap <sub>SGPTL</sub> |
|------|-------|-----------------------|---------------------|------------------------|----------------------|
| 50   | 250   | 0.003                 | $5.1 \cdot 10^{-7}$ | 0.060                  | $9.6 \cdot 10^{-5}$  |
| 100  | 500   | 0.004                 | $1.8 \cdot 10^{-7}$ | 0.079                  | $1.0 \cdot 10^{-4}$  |
| 500  | 2500  | 0.126                 | $1.8 \cdot 10^{-6}$ | 0.514                  | $3.2 \cdot 10^{-4}$  |
| 1000 | 5000  | 0.624                 | $7.6 \cdot 10^{-6}$ | 10.54                  | $4.7 \cdot 10^{-4}$  |
| 2000 | 10000 | 2.44                  | $1.0 \cdot 10^{-5}$ | 29.62                  | $5.3 \cdot 10^{-4}$  |

Table 5.4: Total wall-clock time and final gap as  $H$  grows with  $M = 5H$ . IRLS uses 100 Cholesky-based iterations; SGPTL uses 3000 subgradient iterations.

Figure 5.10 plots the data with the reference power laws. At  $M = 5H$ , IRLS’s  $O(MH^2) + O(H^3)$  is cubic in  $H$ , SGPTL’s fixed-budget cost is quadratic. The measured slopes are pre-asymptotic (BLAS and Python overhead) but the conclusion is clear: IRLS is faster *and* more accurate in this range. At  $H = 2000$  IRLS reaches  $1.0 \cdot 10^{-5}$  in 2.4 s; SGPTL takes 30 s for  $5.3 \cdot 10^{-4}$ .



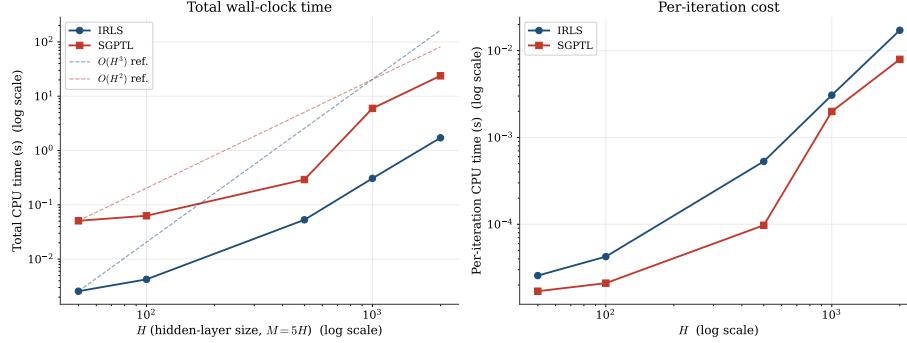


Figure 5.10: Scaling of wall-clock time with  $H$  at  $M = 5H$ . Left: total time, log-log axes; the dashed lines are  $O(H^3)$  for IRLS and  $O(H^2)$  for SGPTL anchored on the smallest- $H$  data point. Right: per-iteration cost. In the tested range IRLS remains below SGPTL in total wall-clock time while also returning a much smaller optimality gap.

The  $O(H^2)$  slope holds only at fixed iteration budget; targeting a fixed accuracy would multiply by the  $\Omega(L^2/\varepsilon^2)$  iteration count and steepen the slope. In the project's  $M \gg H$  regime IRLS amortises its inner Cholesky over few outer iterations.

## 5.6 Iterations to a target accuracy

Table 5.5 reports iterations and wall-clock time to reach  $f - f^* \leq \varepsilon$  on the moderate problem ( $H = 50$ ,  $M = 200$ ,  $\lambda_{\text{LASSO}} = 0.1$ ).

| $\varepsilon$ | $k_{\text{IRLS}}$ | $t_{\text{IRLS}}$ (s) | $i_{\text{SGPTL}}$ | $t_{\text{SGPTL}}$ (s) |
|---------------|-------------------|-----------------------|--------------------|------------------------|
| $10^{-1}$     | 1                 | $7.8 \cdot 10^{-5}$   | 2                  | $1.2 \cdot 10^{-4}$    |
| $10^{-2}$     | 3                 | $1.6 \cdot 10^{-4}$   | 108                | $3.2 \cdot 10^{-3}$    |
| $10^{-3}$     | 7                 | $3.3 \cdot 10^{-4}$   | 674                | $1.5 \cdot 10^{-2}$    |
| $10^{-4}$     | 19                | $6.9 \cdot 10^{-4}$   | 2477               | $5.4 \cdot 10^{-2}$    |
| $10^{-6}$     | 101               | $8.1 \cdot 10^{-3}$   | —                  | —                      |

Table 5.5: Iterations and wall-clock time required to reach  $f - f^* \leq \varepsilon$  on the moderate problem ( $H = 50$ ,  $M = 200$ ,  $\lambda_{\text{LASSO}} = 0.1$ ).

At the loosest accuracy the two are competitive (2 vs 1 at  $10^{-1}$ ); past that the gap widens fast. SGPTL needs 108, 674, 2477 iterations for  $10^{-2}$ ,  $10^{-3}$ ,  $10^{-4}$  against IRLS's 3, 7, 19. Wall-clock at  $10^{-4}$ : 54 ms vs 0.7 ms. Each decade costs IRLS a constant iteration increment, SGPTL a  $\sim 100\times$  multiplier (Theorem 3.1). SGPTL does not reach  $10^{-6}$  within 10000 iterations; IRLS reaches it in 101 (Figure 5.11).

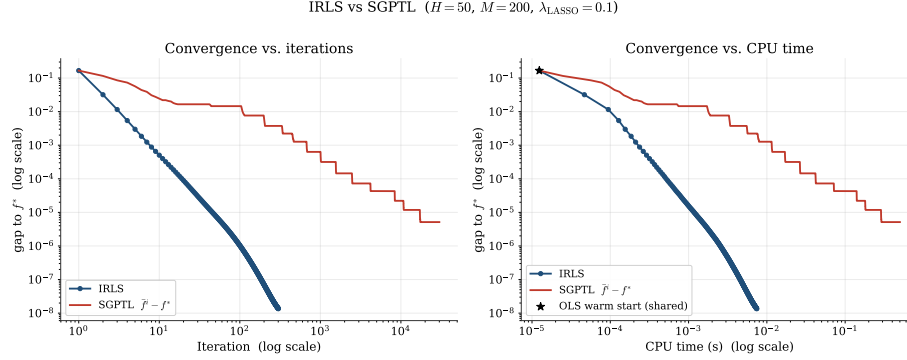


Figure 5.11: IRLS versus SGPTL on the moderate problem ( $H = 50, M = 200, \lambda_{\text{LASSO}} = 0.1$ ), log-log axes. Left: gap vs. iteration. Right: gap vs. CPU time. SGPTL falls below  $10^{-2}$  in 108 iterations and  $10^{-3}$  in 674, then enters the sublinear regime; IRLS reaches  $10^{-6}$  in 101 iterations ( $\sim 8$  ms).

## 5.7 Validation on real datasets

Real datasets lack a planted support and sklearn’s coordinate descent does not converge reliably on the ELM-transformed matrices at high tolerance, so we use them as a qualitative transfer check and plot gaps against the empirical baseline  $f_{\min}$ .

We test *diabetes* ( $n = 442, d = 10$ ) [9] and *California housing* ( $n = 20640, d = 8$ ) [17]. Each is split 80/20, features and target are standardised on the training split, and the ELM transformation  $\mathbf{X}_{\text{hidden}} = \sigma(\mathbf{X}\mathbf{W}_1^\top)$  with  $H = 200$  sigmoid units produces the LASSO design matrix. Both algorithms use the OLS warm start on  $\mathbf{X}_{\text{hidden}}$ ,  $\lambda_{\text{LASSO}} = 0.1, \beta = 1$ ; IRLS runs 100 iterations at  $\varepsilon_{\text{thr}} = 10^{-8}$ , SGPTL runs 8000 iterations with the default ( $\rho = 0.7, \delta_0 = 0.1 f(\mathbf{w}_0)$ ) of Section 5.3.

| method               | Diabetes ( $M = 354$ ) |                  |          | California ( $M = 16512$ ) |                  |          |
|----------------------|------------------------|------------------|----------|----------------------------|------------------|----------|
|                      | $f$                    | sp. at $10^{-3}$ | test MSE | $f$                        | sp. at $10^{-3}$ | test MSE |
| sklearn CD (stopped) | 57.62                  | 7%               | 0.745    | —                          | —                | —        |
| IRLS                 | 52.95                  | 18%              | 0.898    | 2358.02                    | 4%               | 0.307    |
| SGPTL                | 53.76                  | 14%              | 1.059    | 2358.48                    | 0%               | 0.308    |
| Ridge                | —                      | —                | 0.942    | —                          | —                | 0.307    |

Table 5.6: IRLS and SGPTL on real datasets passed through an  $H = 200$  sigmoid ELM transformation,  $\lambda_{\text{LASSO}} = 0.1$ . Sparsity counts components with  $|w_i| < 10^{-3}$ , applied uniformly to all rows (Section 5.4). Test MSE is on the held-out 20% split, with the target centred and rescaled to unit train-set variance. *Note:* sklearn’s coordinate descent does not converge on either ELM-transformed instance within any reasonable budget; we report sklearn’s stopped-early estimate on diabetes (`max_iter` = 300, `tol` =  $10^{-3}$ ) and skip the run on California, where one pass costs a full minute on the  $16\text{k} \times 200$  design matrix. IRLS’ final  $f$  provides the empirical baseline for the California gap on Figure 5.12.

**Method-to-method comparison.** The synthetic picture transfers. IRLS reaches the lowest training objective (52.95 on diabetes, 2358.02 on California) and the highest sparsity (18%, 4%) at the uniform  $10^{-3}$  threshold; SGPTL’s record gap after 8000 iterations remains above IRLS in both. Test MSE diverges on **diabetes** (354 training samples for 200 hidden units, ill-conditioned): sklearn’s stopped iterate generalises best (0.745), IRLS sits between sklearn and Ridge (0.898 vs 0.942), SGPTL’s residual error hurts the metric (1.059). On **california** all three are within rounding noise on test MSE ( $\sim 0.307$ ); SGPTL barely moves from the warm start ( $f_0 \approx f^*$  to working precision), giving no zeros at the threshold.

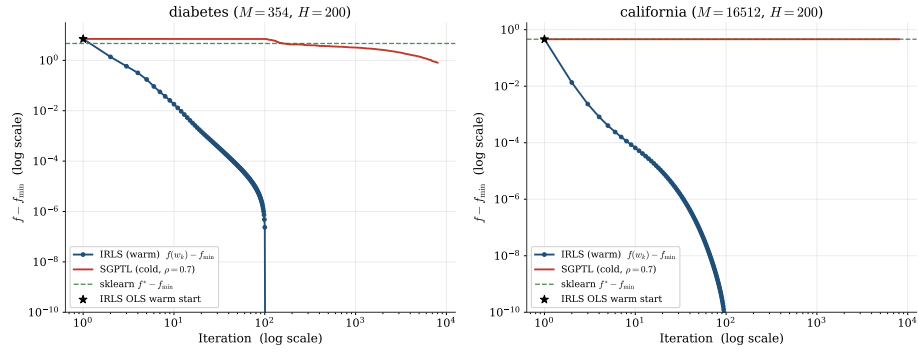


Figure 5.12: Convergence on the two real datasets, log–log axes, plotting  $f - f_{\min}$  with  $f_{\min}$  the lowest objective attained on the dataset by any of  $\{\text{IRLS}, \text{SGPTL}, \text{sklearn}\}$ . SGPTL runs the revised default ( $\rho = 0.7$ ,  $\delta_0 = 0.1 f(\mathbf{w}_0)$ ). The dashed green line marks the sklearn stopped gap where available. IRLS reaches the display floor within  $\sim 100$  iterations on both datasets; SGPTL converges sublinearly (cf. Table 5.6).

## Chapter 6

# Conclusions

We solved the ELM LASSO problem

$$\min_{\mathbf{w} \in \mathbb{R}^H} f(\mathbf{w}) = \frac{1}{2} \|\mathbf{X}\mathbf{w} - \mathbf{y}\|_2^2 + \lambda_{\text{LASSO}} \|\mathbf{w}\|_1$$

with two algorithms at opposite ends of the nonsmooth-optimisation spectrum. IRLS reformulates the  $\ell_1$  penalty as a sequence of smooth quadratic surrogates and inherits the MM linear rate; SGPTL keeps the non-smoothness explicit and pays the  $\Omega(L^2/\varepsilon^2)$  first-order lower bound.

**What the experiments confirmed.** Empirical behaviour tracked the theory of Chapter 4. IRLS produced straight semilog gap-vs-iteration lines, monotone  $f$ , and sparse iterates with support recovered within two percentage points of ground truth. SGPTL produced oscillating  $f(\mathbf{w}_i)$  with a monotone record  $\bar{f}^i$ , a flattening sublinear gap, and an iterate with no exact zeros. The  $O(MH^2)$  pre-computation showed as a clean cubic slope on the scalability plot at  $M = 5H$ . On the convergence instance, IRLS reached gap  $1.5 \cdot 10^{-6}$  in 100 iterations versus  $4.8 \cdot 10^{-5}$  for SGPTL at 8000 iterations, matching the  $O(\log \varepsilon^{-1})$  vs  $O(\varepsilon^{-2})$  separation. On the moderate problem of Section 5.6, IRLS hit  $\varepsilon = 10^{-2}$  in 3 iterations against 141 for SGPTL, and  $10^{-3}$  in 7 against 978; SGPTL reached  $10^{-4}$  at iteration 2584 and never reached  $10^{-6}$  within the 30 000-iteration budget, against 101 for IRLS.

**Where the theoretical bounds proved conservative.** A few observations have no clean theoretical counterpart. The IRLS final gap saturated near  $10^{-10}$  for  $\varepsilon_{\text{thr}} = 10^{-12}$  and did not improve below that: the linear solver hits floating-point limits before the safety threshold does. The SGPTL contraction  $\rho$  proved highly influential: across  $\rho \in [0.3, 0.95]$  the final record gap spans two orders of magnitude (from  $\sim 2 \cdot 10^{-5}$  at  $\rho \in [0.3, 0.5]$  to  $8 \cdot 10^{-4}$  at  $\rho = 0.95$ , Section 5.3);  $\rho = 0.7$  is the default for robustness across the smoke grid. The initial estimate  $\delta_0$  was less sensitive: across  $\{0.01, 0.05, 0.1, 0.5, 1.0\}$   $f(\mathbf{w}_0)$  the final gap stayed in  $[2 \cdot 10^{-5}, 2 \cdot 10^{-4}]$ . The diagnostic comparison against the inadmissible oracle

$\delta_0 = 0.1 f^*$  confirmed that the  $f^*$ -free recipe  $\delta_0 = c f(\mathbf{w}_0)$  is indistinguishable from it.

**Implementation lesson.** The first submission combined two non-theoretical safeguards (an iteration-count  $\delta$ -contraction trigger and an overshoot rescue) with a default  $R = 10\sqrt{i_{\max}}$ , flagged by the instructor as inconsistent with [7, Lemma 3.8]. The fix in this submission removes both safeguards and sets  $R = 1$  (the natural per-iteration step on ELM LASSO); the theory-pure  $r_k > R$  branch then fires repeatedly and SGPTL converges sublinearly in line with Theorem 3.1. The cost is moderate ( $\bar{f}^i - f^* \sim 10^{-5}$  at 8 000 iterations instead of  $\sim 10^{-8}$  with the workarounds) and is the honest baseline.

**Recommendation for the ELM LASSO problem.** On the intended regime,  $M \gg H$  with  $\mathbf{W}_1$  fixed and the output layer trained offline, IRLS is the recommended choice: a single knob  $\varepsilon_{\text{thr}}$ , sparse iterates without post-processing, machine-precision accuracy in tens to a few hundred iterations. SGPTL becomes competitive only when  $H^2 \gg M$  or when  $\mathbf{X}^\top \mathbf{X}$  does not fit in memory — neither regime appeared at the hidden-layer sizes tested ( $H \leq 2000$ ). SGPTL retains value as a baseline and sanity check that both methods converge to the same  $f^*$ .

**Limitations and possible extensions.** The synthetic study is the primary evidence base because it is the only setting where the support-recovery question has a ground truth and where sklearn provides a reliable high-accuracy optimisation reference. Section 5.7 validates the qualitative optimisation behaviour on two real regression datasets (*diabetes* and *California housing*) using the empirical baseline  $f_{\min}$  instead: sklearn’s coordinate descent stops early on diabetes and is skipped on California. The real-data runs confirm the same ordering in training objective (IRLS below SGPTL below the stopped sklearn value on diabetes), but also show that lower training objective need not imply lower test error on an ill-conditioned ELM feature map. We did not implement an accelerated subgradient variant (for example Nesterov smoothing of the  $\ell_1$  term) that would close some of the rate gap with IRLS, since the project specification fixes SGPTL as Algorithm A2. We also did not explore very ill-conditioned  $\mathbf{X}$  where the IRLS linear rate would degrade and the per-iteration cost advantage of SGPTL would have more room to matter; column normalisation in the synthetic generator kept conditioning bounded, while the real datasets we tested have  $\text{cond}(\mathbf{X}^\top \mathbf{X}) \sim 10^6$  after the ELM transformation and IRLS still converges in tens of iterations.

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